

Beyond Getting Work Done: A Landscape Review of Semantic Scaffolding for Learning with AI

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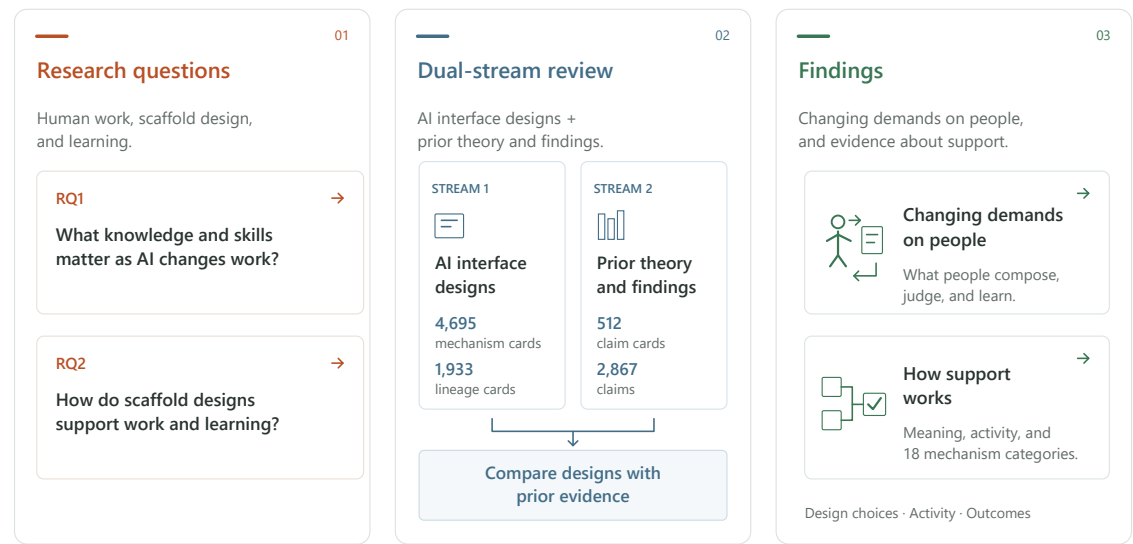


Fig. 1. **The review at a glance.** Two research questions guide a dual-stream review of AI interface designs and prior theory and findings. The findings examine how work changes and how support is designed and evaluated. Card counts cover the selected AI papers in Stream 1 and retained sources in Stream 2.

ABSTRACT

As AI changes what people can delegate, it also changes what they have opportunities to understand, practice, and learn. We examine how interfaces can support development within useful work through *semantic scaffolding*: designed structures that make task meaning available for human reasoning and action. Our dual-stream landscape review connects 4,638 AI-involved papers from 40 venues (2019–2026) with 512 sources on prior theory and empirical findings. GenAI papers more often delegate generation while supporting people in creating, editing, and repairing artifacts. The examples reveal demands on interpreting material, judging its grounds, and making reasoning usable by others. Eighteen mechanism categories map how represented meaning connects to human activity. Their comparisons show why supplying material can create practice as well as remove it, and why what a representation makes inspectable matters to the judgment performed. Evaluations also separate better work from learning: CodeVoyager improved immediate closed-book comprehension without a completion advantage, while required reflection reduced completed practice from 3.4 to 2.0 problems within a fixed time. We argue for choosing capabilities with people, designing support that evolves with their work and development, and connecting evidence about process, artifacts, and learning. The review provides concrete design alternatives and a research agenda for learning within everyday AI tools.

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CCS Concepts: • **Human-centered computing** → **Empirical studies in interaction design**; *Graphical user interfaces*; *Collaborative interaction*; Natural language interfaces.

Additional Key Words and Phrases: Semantic scaffolding, learning, human–AI interaction, external cognition, landscape review, interface mechanisms

1 INTRODUCTION

AI tools help people write, program, analyze information, and make decisions. They can supply material people work with: alternative passages, intermediate plans, explanations, and visual representations [63, 72, 154]. Interfaces let people inspect, steer, and revise this material as a task develops. A writer can work from generated alternatives; a programmer can inspect a proposed solution; a team can reason through a shared model. What people need to know changes with what these tools make possible. This raises a design question beyond completing the current task: what should people have opportunities to understand, practice, and learn?

We take the position that people should be able to develop through their work even as tools become more capable. The capability worth developing is a moving target: producing an artifact, judging alternatives, and directing delegated work can matter differently to different people. Helpful output alone gives little guidance about which of these to practice. The assistance dilemma captures part of this tension: support can enable progress while taking over an activity the learner needs to practice [73]. Learning depends on how people engage with the material, not simply on how much interaction an interface requires [28]. A critical review of HCI research also finds that claims about critical thinking often outrun clear definitions and assessments [62]. Our review finds a related gap in evidence over time: among 396 AI-involved sources in the selected outcome appraisal, only 31 assessed capability in a later session. We therefore need to connect the work an interface makes possible with opportunities to develop and evidence of what carries into later work.

Semantic scaffolding is our account of how designed structures make task meaning available for people to reason and act with. A concept map relates ideas; a comparison makes differences available for judgment; an editable plan connects actions to a goal. Their learning potential lies in what people do through them: explaining a relation, choosing a criterion, or revising a plan in response to evidence. This gives support a dual role: helping someone make progress now and creating opportunities to develop understanding, skill, and judgment through that work. Scaffolding theory explains responsive support that fades as responsibility passes to the learner [141, 151]; cognitive-artifact research explains representations that remain useful as people become more skilled [107, 115]. Generative and malleable interfaces make it possible to create, reshape, and preserve these structures during use [23, 67, 100]. The design challenge is to make support fit both the decision at hand and the person’s development: some assistance can fade, while useful structures persist or evolve. Section 3 develops this account.

We investigate this challenge through an LLM-assisted, dual-stream landscape review. Language models processed the papers into structured evidence cards with supporting passages and page references. Stream 1 screened 35,480 abstract-bearing records from 40 venues spanning 2019–2026, selecting 7,372 for full-text processing. The current card corpus contains 4,638 AI-involved papers. Inclusion requires AI involvement, an implemented or studied structure with task meaning, and a person who inspects or acts through it. Stream 2 starts from 795 candidate works, of which 543 have full-text claim cards, and contributes 512 retained sources on prior theory and empirical findings, represented by 2,867 claims. We connect the designs to human activity, their theoretical rationale, and evidence about performance and learning. We ask:

- **RQ1.** What knowledge and skills matter as AI changes work?
- **RQ2.** How do scaffold designs support work and learning?

We approach RQ1 through the workflows and cognitive work that the reviewed designs support. GenAI papers more often delegate generation to the tool while also supporting people in creating, editing, and repairing artifacts. The examples reveal demands on knowledge: interpreting generated material, judging its grounds, and making one's reasoning usable by another person. These suggest learning priorities to consider alongside people's purposes. For RQ2, 18 mechanism categories distinguish how interfaces make task meaning available and how people work with it. The comparisons show why an account of learning needs both: CodeVoyager improved immediate comprehension without a detected advantage in assisted completion, while required reflection used practice time without establishing a transfer advantage [7, 70]. These findings motivate support that creates opportunities for meaningful practice and changes as people develop, while retaining resources useful to their work.

This paper contributes:

- a landscape of changing human work with AI, connecting quantitative technology comparisons to the knowledge and judgment that particular designs call for;
- an illustrated account of semantic-scaffolding mechanisms that connects represented task meaning, human activity, and concrete examples;
- an evidence-grounded position on learning within everyday tools, linking design choices and prior theory to process, artifact, and learning outcomes and to questions for future systems and evaluations.

2 BACKGROUND

Research on AI assistance explains ways to distribute work between people and tools. Learning research asks how people develop through that activity, and interface research offers ways to change its representations. Bringing these accounts together helps us ask which opportunities for development belong within tools for everyday work.

2.1 Assisting Work and Supporting Learning

Distributed cognition follows work across people, tools, and representations [52]. External resources can become part of how people remember and reason [32, 61]. Offloading also changes what people retain: expected access to information shifts memory toward where to find it [134], while automation can leave operators with the hardest cases and little practice [8]. These accounts motivate examining the activity left to the person alongside the output of the person-tool system.

AI interaction research makes that division of work a design choice. Mixed-initiative interfaces combine automation and direct manipulation [56]; interactive machine learning centers the user-system loop [3]. Explainable-AI frameworks separate understanding, trust, debugging, and decision support [105], while teaming and learning-support research adds capability and transfer [22, 49]. Generative systems expose prompts, plans, alternatives, and evidence for people to inspect and steer [24, 72, 137, 153]. When systems also supply criteria and explanations, people may need to judge the guidance as well as the answer.

Learning research explains why reducing effort is an incomplete design goal. The assistance dilemma asks when giving or withholding help enables learning [73]; ICAP distinguishes the engagement an activity invites [28]. Guidance useful to novices can become redundant with expertise [65], and difficulty during practice can support later retention

[15]. People also trade accuracy against effort [114]. Together, these accounts direct attention to which work assistance removes, which work remains worth doing, and how that balance changes as a person develops.

2.2 Interfaces People Can Inspect and Reshape

Recent design spaces make representational choices explicit. Min et al. separate an abstraction's schema from its representation and examine granularity, presentation, and transformation [101]. Kim et al. organize how generative interfaces elicit preferences along six axes, including when a request appears and which part of an artifact it concerns [68]. These accounts show how interface structure can help people express and work with task knowledge.

Interfaces can adapt controls to a person's abilities [45, 149]. Generative systems extend what can be constructed during use [23]: Google's Generative UI work creates interactive pages and simulations in response to a prompt [81, 82]. Malleable overview-detail interfaces let people change the content, composition, and layout of views [100]; Meridian provides a specification and tools for building those transformations [102]. Maru preserves a user's information-architecture rules across generation [67]. Generating, reshaping, and preserving support offer different ways an interface might respond to changing work and developing expertise.

2.3 What Prior Reviews Establish and Leave Open

Scaffolding syntheses connect contingent support and learner activity to learning [11, 141]. Self-explanation research identifies practice exposure and redundant prompting as consequential conditions [122]. Krakauer distinguishes artifacts whose advantage remains only during use from those that leave capability behind [74]. Inie et al.'s critical scoping review examines how HCI research about AI defines and assesses critical thinking [62]. This work supplies both substantive learning principles and reasons to examine whether a study's measure tests its proposed benefit.

HCI landscapes offer complementary ways to compare designs. Creativity support reviews relate tools to practice and evaluation [42], organize traces of creative activity [51], and examine generative tools in artists' work [72]. Design spaces organize recurring choices and their trade-offs [94]; their visual form shapes what designers explore [14, 79]. Human-AI guidelines already address intelligibility and correction [4]. Citeology, Royal Constellations, and HCI Research Trails make relationships among cited works navigable [19, 98, 104].

These bodies of work supply different parts of the design problem. Learning theory explains why an activity might develop a capability; interface research shows ways to make that activity possible; evaluations establish what happened in a particular setting. Our review connects these parts across domains. An explanation, an editable plan, and a shared view can all support reasoning, but leave different work to the person. The next section develops semantic scaffolding as an account of that relationship and its potential for learning.

3 WHAT IS SEMANTIC SCAFFOLDING?

A tool changes both what a person can accomplish and what they have occasion to learn. These changes need not move together. A plan can help someone finish an unfamiliar task; constructing or revising that plan can help them understand how its steps fit together. Which support is useful depends on the work, what the person already understands, and what they want to become able to do. We develop semantic scaffolding from this relationship, asking what earlier accounts explain, what generative AI changes, and what should persist as the work itself changes.

3.1 What Is Old: Tools Change the Work of Thinking

Cognitive artifacts reorganize a task by storing, displaying, or transforming information [107]. A checklist turns remembering into preparing, consulting, and interpreting a list. A calculator performs arithmetic while the person formulates a problem and judges its result. Such tools change the knowledge needed to work well with them. External-representation research explains why even the same information supports different operations when arranged differently [126, 160]. People can also change their surroundings to aid reasoning and coordinate several representations as they work [2, 71]. The design question is therefore which thinking a structure performs and which thinking it makes possible for the person.

That relationship develops through use. Wendy Mackay’s account of *co-adaptation* describes people repurposing email filters from sorting messages before reading to filing them afterwards, and developers responding to that practice [93]. People learn ways of working with a tool and change its role in their work. A scaffold designed around an initial task or an assumed novice can consequently outlast the need that motivated it, acquire a new purpose, or need to be changed.

McLuhan’s hot and cool media distinguish the detail a medium supplies and the participation it asks of an audience [99]. Our proposed hot/cold information lens extends this idea to *supplied inference*: how much interpretation a representation provides and what its user works out. Observations, a structured comparison, and a recommendation may concern the same problem while doing different parts of the reasoning for the reader. This extension helps articulate a design choice left open by visual form alone. Supplying an inference can free attention for another judgment; withholding it can create an occasion to reason. Its learning value depends on which activity the person undertakes.

3.2 What Is New: The Support Can Be Generated and Reworked

Generative AI can supply both material for a task and structures for working with it. AI Chains exposes intermediate steps that people can inspect and change [154]; Luminate supplies dimensions for comparing ideas; VIVID generates rubrics and expected answers alongside candidate dialogues [30, 137]. Design rationale has long connected questions, options, and criteria [94]. Generating these together places a consequential choice in ordinary use: a system can help formulate the problem and its standards as well as propose an answer. A person may need to judge the plan or criterion through which the answer was produced, then decide what to preserve or change.

The representation can also change during the work. Generative interfaces create interactive material; malleable interfaces let people reshape views; and persistent information architectures retain organizing rules across generation [23, 67, 100, 102]. Together these possibilities widen the choices for support: what it supplies, who shapes it, and when it changes. Fitting a new task and responding to a person’s development become related design problems. As a person learns to plan, for example, a tool could shift from supplying steps to helping them compare and revise their own plans, while continuing to organize a difficult project. Changing the interface makes such designs possible; responding well requires understanding the person’s goals, knowledge, and experience.

3.3 What Persists: People Need Opportunities to Develop

The purpose of scaffolding is to make meaningful participation possible. Wood, Bruner, and Ross describe a tutor taking over parts of a task beyond a learner’s current capacity so that the learner can undertake the rest [151]. Developmental scaffolding responds to the learner, reduces assistance as capability grows, and transfers responsibility [141]. Reiser distinguishes structuring a task from making its important problems available for reasoning [120]; Quintana et al.

develop these concerns into supports for inquiry [119]. These accounts make the activity worth undertaking the starting point for design. An easier task and a productive opportunity to learn can require different kinds of help.

Reflection, comparison, and analogy offer enduring ways to engage with that activity. Schön’s reflective practice connects action to reconsidering a situation in light of its response [128]. Structural comparison can make relevant differences noticeable [48]; analogy brings relationships from a familiar situation to a new one [47]; variation theory asks which distinctions a learner needs to discern [96]. These approaches inform both narrowing and opening possibilities: a contrast can help someone discriminate between alternatives, while an analogy can suggest a different way to frame the problem. A tool can supply examples, organize the comparison, or prompt a prediction while leaving the target reasoning available to the person.

What deserves that practice changes with people’s purposes and their tools. A writer may want to develop prose, editorial judgment, or both. Choosing one need not make the other obsolete. The persistent commitment is to people having opportunities to develop capabilities they value and can use later. Those capabilities may include working more effectively with continuing resources. Pea distinguishes developmental scaffolding that fades from continuing distributed intelligence [115]; useful support can remain distributed across people and tools [10]. We therefore need to specify what a person is becoming able to do, and with which resources, rather than making the removal of every tool the endpoint of learning.

3.4 Our Position: Design the Relationship between Meaning and Activity

We use *semantic scaffolding* for support that makes task meaning available through a designed structure a person can reason or act with. The meaning may be a relationship between concepts, a criterion for a choice, an intermediate step toward a goal, or evidence for a claim. The scaffold is the relationship between that structure and the person’s activity. A concept map, for example, gives form to relations; exploring, explaining, or revising those relations are different uses of it. Graphologue lets people explore concepts and their connections [63]. Zong et al. use the term more specifically for explanations of domain groupings to blind and low-vision readers [165]. Our account connects these forms through the task meaning they provide and the activity they support.

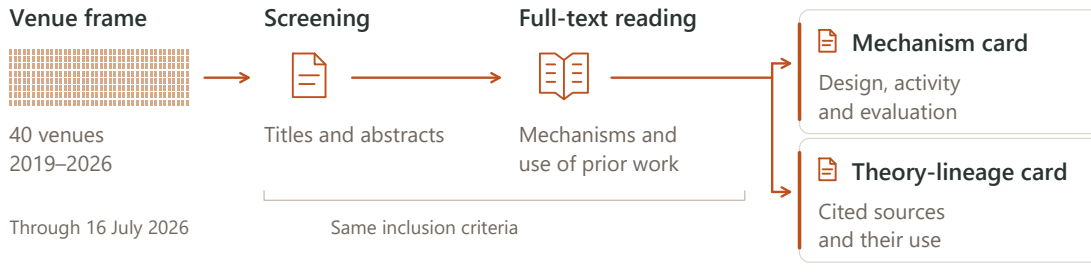
This focus complements affordances and signifiers, which concern available actions and how people discover them [108]. HaLLMark’s highlight both signals authorship and connects a passage to its production history [55]. In Codestrates v2, a tree both organized code and governed execution order, a consequence a participant discovered during use [18]. Learning to operate a control and understanding the task relationship it exposes are connected but distinguishable achievements.

For learning, the design problem is how to use this relationship to help someone develop through the work. A person might follow a supplied plan, explain why a step is needed, or construct a plan with feedback. The representation, the activity, and the opportunity to learn differ even when the final task is the same. The two roles of support—accomplishing work and developing capability—can coexist in one tool and change over time. Prompts may fade, a useful representation may remain, and the work a person chooses to undertake may grow more demanding. The broad corpus criteria in Section 4.1.3 let us examine those possibilities without assuming that every included design has already demonstrated learning. This preserves the developmental question raised by scaffolding research [115, 118, 131].

Our position is that interfaces should make room for this development within useful work. To inform their design, we connect the meaning a structure makes available, the activity documented in its use, and the consequence assessed. The review asks what knowledge and skills that work calls for, how mechanisms support it, and which proposed learning opportunities the evaluations support, challenge, or leave untested.

Literature selection and evidence extraction

Stream 1 · AI-involved designs



Stream 2 · Prior theory and findings

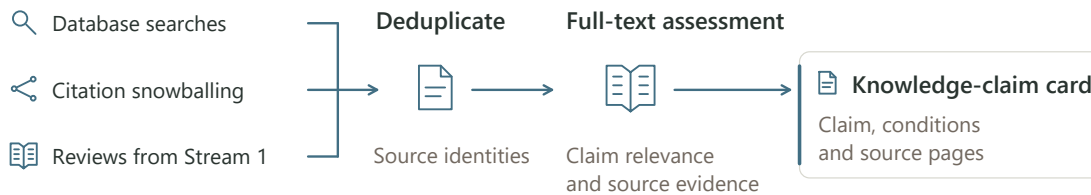


Fig. 2. **Literature selection and evidence extraction.** Stream 1 runs from a venue frame through screening and full-text reading to mechanism and theory-lineage cards. Stream 2 combines three discovery routes, removes duplicates, and records each source's claims on a knowledge-claim card.

4 METHODS

We reviewed two bodies of work (Figure 2). Stream 1 is a systematic corpus of AI-involved systems and studies from fixed venues and years; it supplies the designs we map. Stream 2 collects theories and empirical studies of how support works, including studies without AI; we interpret the designs against it. The two are counted separately. Scoping-review guidance shaped selection, extraction, and reporting [6, 80, 139], and earlier HCI landscape reviews shaped the venue frame and synthesis [42, 51].

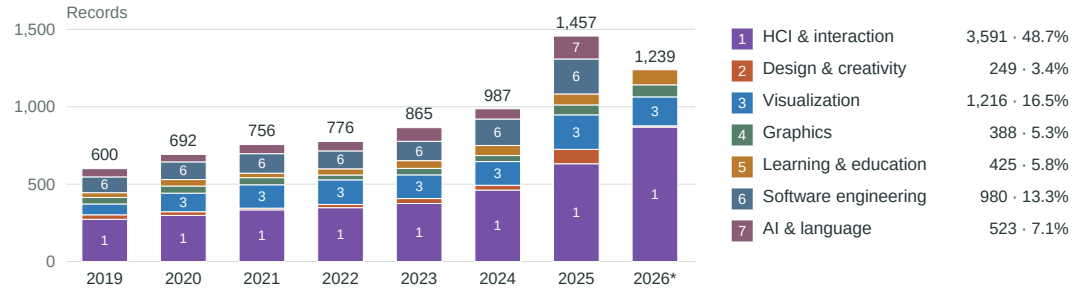
4.1 Literature Review

4.1.1 Venues. Stream 1 covers 40 venues in HCI, design and creativity, visualization, graphics, learning and education, software engineering, and AI and language, published from 2019 to 2026 (Figure 3). Papers are dated by their version of record; the selection figure, the design space, and the trend figures cover papers published by 16 July 2026. Ineligible publication types were excluded by rule, and citation counts played no part.

4.1.2 Screening. Two language models separately screened the 35,480 records with an abstract; the 2,498 without one were not screened. Their verdicts, a review of their disagreements, and (for the first reading list) PDF availability determined the 7,372 records selected for full-text reading. One language model processed each full text and recorded whether it met the criteria, with supporting passages and page references.

Selected records by year and venue group

7,372 records selected for full-text reading; venue groups describe disciplines, not application domains.



Each venue: selected records and share of the 7,372 selections

1 CHI	1,548 · 21.0%	1 ECSCW	10 · 0.1%	2 Design Studies	13 · 0.2%	5 AIED	58 · 0.8%
1 UIST	306 · 4.2%	1 GROUP	6 · 0.1%	2 IDETC	151 · 2.0%	5 IJAIED	37 · 0.5%
1 IUI	359 · 4.9%	1 ASSETS	84 · 1.1%	3 VIS	394 · 5.3%	5 EDM	103 · 1.4%
1 DIS	232 · 3.1%	1 HRI	187 · 2.5%	3 EuroVis	111 · 1.5%	6 CHASE	55 · 0.7%
1 CSCW	258 · 3.5%	1 IVA	82 · 1.1%	3 TVCG	635 · 8.6%	6 ICSE	321 · 4.4%
1 C&C	67 · 0.9%	1 CHIIR	56 · 0.8%	3 VL/HCC	76 · 1.0%	6 FSE	238 · 3.2%
1 TOCHI	45 · 0.6%	1 HCOMP	36 · 0.5%	4 SIGGRAPH	295 · 4.0%	6 ASE	366 · 5.0%
1 TiIS	102 · 1.4%	1 IJHCS	127 · 1.7%	4 SIGGRAPH Asia	93 · 1.3%	7 ACL	191 · 2.6%
1 EICS	33 · 0.4%	2 ICCS	53 · 0.7%	5 L@S	41 · 0.6%	7 EMNLP	255 · 3.5%
1 HAI	53 · 0.7%	2 Design Science	32 · 0.4%	5 LAK	186 · 2.5%	7 RecSys	77 · 1.0%

*2026 is partial: publication cutoff 16 July. Zero selections can reflect unavailable proceedings.

Fig. 3. **Records selected for full-text reading, by publication year and venue group.** The lower panel gives each venue's count and share of the 7,372 selections. The 2026 bar covers papers published by 16 July.

The AI requirement was added partway through full-text reading, so three separate model passes rechecked the 1,154 earlier cards that left the question open. A paper left the corpus only when all three found no AI, with low uncertainty and a quoted passage (492 cards). The corpus holds 4,638 papers, and 1,076 of them form a learning subset: their support has a documented learning purpose, or the study examines its learning consequences. Another 475 papers without AI form a labeled comparison set outside the systematic corpus counts; the text marks any it uses as an example.

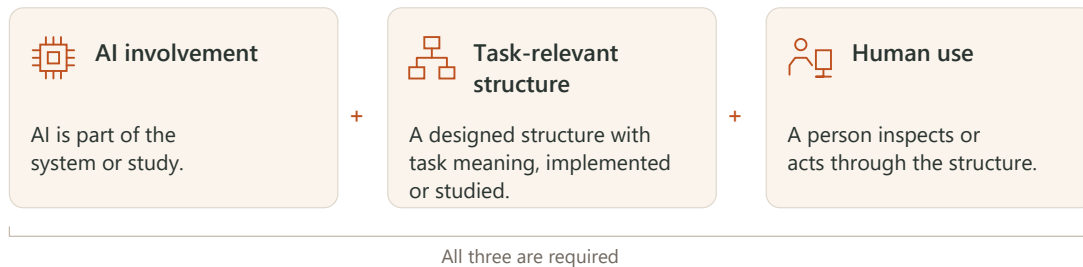
4.1.3 Inclusion Criteria. Stream 1 includes a paper when three conditions hold, at screening and at full text (Figure 4): AI is part of the system or the study; the paper implements or studies a designed structure with task meaning; and a person inspects or acts through that structure. AI is part of a paper when an AI system is built, used, evaluated, or simulated (including Wizard-of-Oz systems presented as AI), or when AI is what the study designs, explains, or labels. Rewording a chatbot's replies (hedges, tone, length) is not a designed structure. Workshop templates, cards, storyboards, and probes used to elicit views count only when the paper builds and evaluates them as its contribution. Reviews, surveys, and meta-analyses go to Stream 2. A design can qualify without a user study; its benefits then count as intended uses, not measured effects. Learning purpose, measured outcomes, adaptation, fading, and the transfer of responsibility are recorded but do not decide inclusion.

4.1.4 Theories and Empirical Studies. Stream 2 gathers theories, design principles, and empirical findings about how support works. Its sources came from database searches, backward and forward citation snowballing [150], and reviews found in Stream 1, and every source received the same full-text assessment of relevance and support. Of 795 candidate works, language models processed 543 full texts into claim cards; the model-assisted assessment excluded 31 for

Inclusion criteria

Stream 1

Title/abstract screening → Full-text reading



Stream 2 · Prior theory and findings

Relevant, transferable claim → Full-text support and conditions

Fig. 4. **Inclusion criteria.** Stream 1 requires all three conditions at screening and full text. Stream 2 assesses whether a claim is relevant and supported by its full text. Learning purpose and measured outcomes are recorded during extraction.

recorded reasons, such as falling outside the review questions or appearing in a questionable venue. The analysis uses the remaining 512 sources and their 2,867 claims. Books, foundational theory, null findings, and studies without AI are eligible.

4.2 Analysis

RQ1 asks what knowledge and skills matter as AI changes work. We approach this through the activities interfaces support, the operations assigned to tools and people, and their variation by technology, year, and context. Quantitative comparisons locate changes in the literature; examples explain the knowledge and judgment involved in performing the supported work. Together they identify candidate learning priorities to consider in relation to people's goals. This is an interpretation of documented designs and use, not a measurement of society-wide skill value. RQ2 asks how scaffold designs support work and learning. We connect represented task meaning to the activity a mechanism makes possible, then examine its theoretical rationale and the process, artifact, and learning outcomes assessed by studies.

4.2.1 What We Recorded. A language model processed each full text into a structured evidence card, quoting the passages and pages behind its entries.¹ Scientific-document benchmarks and expert-assisted literature mining offer precedents for such passage-linked records [142, 147]. The Stream 1 unit is a *mechanism*, an operation involving a designed structure with task meaning: AI Chains' workflow overview and its step editor are two mechanisms [154]. For each mechanism, the card records what the interface represents, how and when it presents that meaning, and what people can do. It also records how the work changed, what the tool took over and what the person kept, and how the design was evaluated: comparison, measures, duration, participants, and results. A second card records how the paper uses each cited source: to frame a problem, explain a mechanism, motivate the design, inform a measure, critique an

¹Evidence cards: mechanism, theory-lineage, and knowledge-claim cards .

approach, or in passing. The Stream 2 unit is a *knowledge claim*, a finding or theoretical argument about how support works, for whom, and when; each card holds one to eight claims with their evidence, conditions, and limits. Missing information is recorded as not reported or unclear, and author claims, measured results, and reviewer inferences stay apart (fields in Appendix A).

4.2.2 Building the Design Space. Closed fields take predefined values, such as whether learning was measured and for how long; open fields describe mechanisms and human work in each paper’s terms. We merged the 142,877 open-field observations into 1,290 codes and grouped these into 223 categories and 92 families across 18 fields (Appendix A.3). Each grouping links to its observations, so the grouping and the assignments can be checked separately [58]. The mechanism field’s hierarchy is the design space: 5 families and 18 categories (Section 6; codebook in Appendix B). For the four-function view (Figure 12), we assigned each of the 110 mechanism codes to the functions it serves; exploring alternatives serves both representing and evaluating.

We built the vocabulary from the cards completed by 28 August 2026 and coded later cards with it. The design-space counts describe the 2,847 papers of that first set still in the corpus (2,891 for the closed fields); the annual and technology figures draw on the 4,133 papers with usable coding (Appendix A.6). A paper counts once per category, however many mechanisms it has. We classified each paper’s AI as generative or traditional AI/ML by the technology it used, not its year, and adjusted technology comparisons for year and venue, since venue mix alone can create differences (Appendix D).

4.2.3 Comparing Designs and Studies. We used the codes to find cases and the source passages to compare them. For each question, we compared the task, the representation supplied, the actions available, the reported use, and the assessed consequence, recording contrary and unclear cases beside supporting ones. We chose examples that fit a mechanism, varying domain and year; citation counts mattered only after fit. We retrieved prior findings and counterexamples through the seven fields coded on Stream 2 claims, such as the form of support and a claim’s conditions. The lineage cards trace how a cited theory or finding entered a design or a measure (Section 6.5); reference lists and citation counts come from Semantic Scholar (Appendix A.9.2).

In the selected lineage comparisons, we examined the connection between the source’s claim, the implemented feature, the activity attributed to the person, and the assessment. This distinguishes a design rationale from evidence that the proposed activity occurred or helped. Unavailable lineage evidence remains unknown, rather than evidence that a design has no theoretical grounding.

For study results, each contrast keeps its population, comparator, outcome, timing, and the assistance available during assessment, and positive, mixed, null, harmful, and unmeasured results stay distinct. A whole-system result is credited to the whole system, and a learning claim counts as tested only when a study measured learning or transfer. We sorted outcomes into process, artifact, and learning (PAL), each assessed by the study or reported by participants (Section 6.6). We did not pool effects across studies (Section 9).

4.2.4 Checking the Coding. Scripts checked the cards’ required fields and matched their quotations against the source text, and quotations the text layer could not match were checked against page images; counts and plots are computed from the saved data. To test the open-code grouping, a separate model pass coded 1,571 extracted observations in six fields with the existing codes hidden. Agreement was low (adjusted Rand index .079 to .445), and the blind pass grouped more coarsely in five of the six fields (Appendix A.4). The mechanism vocabulary gained 12 codes across the 1,754 papers read after it was built (Appendix E).

4.3 AI Use Disclosure

We used language models to screen records, process full texts into evidence cards, code observations from those cards, group codes, develop and critique the cross-case comparisons, write analysis code, and draft and revise text, tables, and figures. Claude Opus 5 and GPT-5.6 Sol screened titles and abstracts. GPT-5.6 Sol, Claude Opus 5, and Claude Opus 5.5 processed full texts into the Stream 1 cards; Claude Opus 5 and GPT-5.6 Sol wrote the Stream 2 cards; Claude Opus 5.5 ran the three-pass AI check. Claude Opus 5 merged labels into codes, and model-assisted review, including Claude Fable 5.1, grouped the codes into categories and families. GPT-6 Sol linked card evidence for the cross-case comparisons, and GPT-6 Astra developed and critiqued them. GPT-6 Astra, working through Codex, drafted the merged manuscript, and Claude Opus 5.5 revised it. Prompts and run records are kept. The authors set the research questions, scope, criteria, and card formats, directed the synthesis, and are responsible for its conclusions; people have not yet checked the coding or the cross-case interpretations (Section 9).

5 FINDINGS: KNOWLEDGE AND SKILLS IN CHANGING WORK

Our first question asks what knowledge and skills matter as AI changes work. The reviewed workflows suggest three connected priorities: composing with generated material, judging a result and the grounds offered for it, and interpreting work shared with others. Quantitative comparisons locate these activities in the literature; concrete designs show what people need to understand and decide within them. These are candidate learning priorities, whose value depends on people's purposes and future work. The comparisons that follow explain their relevance; Section 6 examines designs that support them and the capabilities studies actually assess.

5.1 Composing with generated material

Research on cognitive artifacts distinguishes what a person and tool achieve together from the task left to the person [107]. Creativity-support research likewise examines how tools support activities such as exploring possibilities and developing ideas [42, 72]. The AI designs in our review make this distinction concrete. CorpusStudio retrieves examples and patterns from prior writing for authors to interpret before composing their prose [33]. ABScribe generates alternative passages that authors keep, edit, and compare [121]. The material supplied changes the writer's starting point and the choices available next. Writing knowledge remains relevant to both designs: authors need it to judge which examples or passages fit their purpose. The opportunity to learn shifts with the work they do with that material.

GenAI moves generation to the tool. In 2,119 papers from 2019–2025 with coded delegated work, the tool generates the artifact in 19.3% of GenAI and 1.9% of traditional AI/ML papers (+16.6 points after adjusting for year and venue; 95% CI 12.7 to 20.5), and it more often generates responses and revisions (+9.4). Analysis moves the other way: the tool less often calculates and models (-11.5), evaluates and classifies (-9.7), or compares and finds patterns (-6.7) (Appendix D.2).

The human activity codes show a corresponding emphasis on working with artifacts (Figure 5). Among 2,426 technology-classified papers with activity coding, creating, editing, or repairing artifacts appears in 359 of 1,213 GenAI papers (29.6%) and 183 of 1,213 traditional AI/ML papers (15.1%). Specifying or refining system behavior appears in 37.6% and 30.0%, respectively; comparing alternatives or testing scenarios appears in 11.6% and 20.0%. These pooled shares describe the activities supported by designs across 2019–2026; the delegated-work comparison above separately adjusts for year and venue in 2019–2025. Across these separately coded subsets, GenAI papers more often describe tool generation and human creation, editing, or repair. The group-level patterns motivate examining how designs connect those operations within a task.

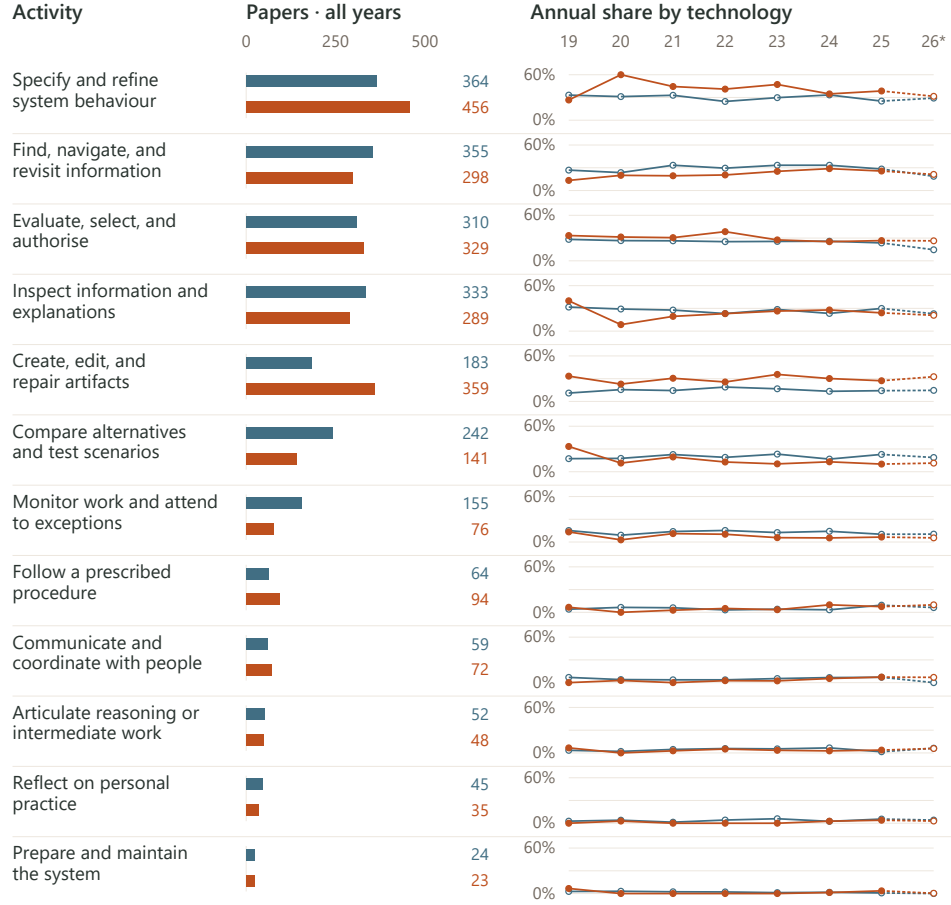
Workflow activities: volume and annual share

2,426 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

■ GenAI

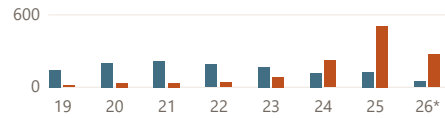
* 2026 partial: through 16 July



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 5. Workflow activities by AI technology. Bars show counts across all years; annual lines show shares within each technology group. Denominators count papers with usable activity codes and a known technology. Activities overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue.

The technology contrast concerns supported operations. The separate domain analysis finds that none of the 16 trends meets the corrected significance threshold after accounting for venue between 2019 and 2025 (Table 12). The quantitative pool contains 4,133 AI-involved papers; each comparison uses those with the required codes. Technology

What does the person do?

Twelve activities that interfaces can support



Fig. 6. What does the person do? Twelve activity groups organize the workflow codes. Each schematic illustrates an action through an interface: finding, inspecting, comparing, judging, directing, creating, articulating, reflecting, monitoring, following a procedure, coordinating, or preparing a system. These activities can occur together; the grid does not imply a sequence. Allocation, supplied support, and measured outcomes are coded separately.

classification follows the system used, including earlier generators and generative hybrids. GenAI accounts for 9.2% of classified papers in 2019 and 80.7% in 2025 (Figure 56). Appendix D gives the annual counts and domain distributions; 2026 covers publication through 16 July.

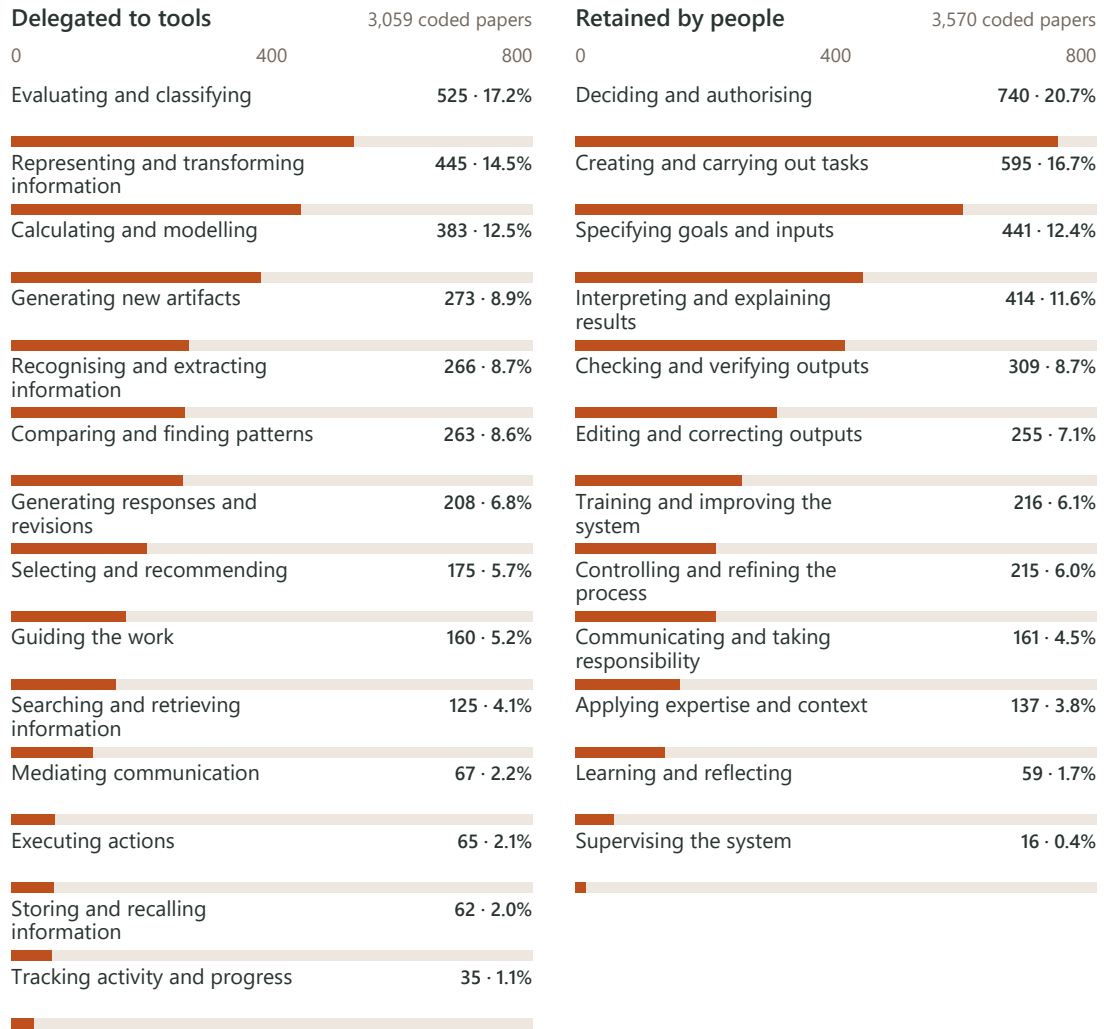
Figure 6 organizes the activities people can carry out with these designs. Activities can recur or combine within a task: an author may inspect examples, specify an intended change, edit a passage, and then reconsider the examples. The map describes kinds of activity; coverage of several kinds alone does not establish support for a complete workflow.

5.2 Judgment needs a basis for checking

Generating material leaves consequential work to the person who will use it. The retained-role codes locate that work: deciding or authorizing appears in 740 of 3,570 papers (20.7%), specifying goals and inputs in 441 (12.4%), and checking or verifying in 309 (8.7%) (Figure 7). The tool and person panels have separate coding denominators and overlapping roles; they describe operations, not amounts of mental effort or improvements in capability.

Work delegated to tools and retained by people

Bar length: number of papers · percentages use each field's coded papers



Counts use all papers coded for each field. Allocation conditions and unspecified work are listed with the data.

Fig. 7. Delegated and retained work, ranked by prevalence. Bar lengths encode paper counts on a common scale; labels also give percentages within each panel's coding denominator. A paper can record several roles. Each panel uses papers with usable codes for that role; technology classification is not required.

Figure 8 shows how these operations fit together. A tool can compose material while a person judges its fit, revises it, or checks its evidence. The design question is which information makes that judgment possible.

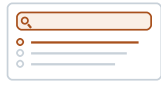
Different references support different checks. Leap exposes values produced when generated code runs [41]; CorpusStudio offers precedents from a writing corpus [33]; ONYX shows how it interpreted an action the user taught it

How is the work divided?

The tool supplies an operation; the person still has work to do.

What the tool does

Perception, search and memory



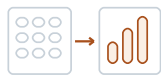
Recognising and extracting information
Searching and retrieving information
Storing and recalling information
Tracking activity and progress

Representation and translation



Representing and transforming information
Comparing and finding patterns

Computation and judgement



Calculating and modelling
Evaluating and classifying

Content generation



Generating new artifacts
Generating responses and revisions

Decision, guidance and action



Guiding the work
Executing actions
Selecting and recommending
Mediating communication

What the person retains

Judgement and decision



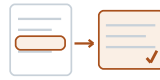
Interpreting and explaining results
Deciding and authorising
Specifying goals and inputs
Applying expertise and context

Doing and revising the work



Creating and carrying out tasks
Editing and correcting outputs
Controlling and refining the process
Training and improving the system

Checking the output



Checking and verifying outputs

Responsibility, social and self-directed work



Communicating and taking responsibility
Learning and reflecting
Supervising the system

Fig. 8. How is the work divided? Five families describe operations supplied by tools; four describe work retained by people. The two columns are independent accounts of a design, not paired transfers. Labels distinguish fourteen tool operations and twelve human activities; learning and reflection are separate from supervising a system. Failed, withheld, absent, and unspecified delegation are recorded separately. Browse examples of tool and human work.

[124]; and Rehearsal simulates how a conversation partner might respond [129]. Execution, precedent, interpretation, and simulation answer different questions. A simulation can support rehearsal without predicting what a particular person will say; a corpus can supply examples without establishing that an idea is new.

Making a reference available is only part of the work. ONYX’s full interface improved script accuracy, yet users missed or wrongly dismissed 44.8% of its follow-up questions [124]. Leap’s live values did not significantly change validation success compared with a condition in which people could still run the code themselves [41]. Together these cases point to a concrete design problem: help people recognize when a check matters, interpret its information, and act on what it reveals. They make source evaluation, domain knowledge, and an understanding of the system’s interpretation relevant to judgment. A verification step is useful only if the person can tell what it establishes about the result.

Generation can also preserve the activity someone needs to practice. ExPeerience uses AI to create problem contexts, tables, and explanations while leaving construction of the SQL solution to the learner [163]. The placement of assistance matters: generating the exercise and generating its solution leave different opportunities to think and act. Section 6 examines the mechanisms that make these choices possible and the evidence for their consequences.

5.3 Who can see the work changes who can help

Distributed-cognition and scaffolding accounts follow support across people and representations [52, 115]. The reviewed designs show why this matters: the person using an interface, the person whose activity it represents, and the person expected to benefit can be different. A useful handoff carries information that someone else can interpret and respond to.

In entrepreneurship coaching, novices describe a project and select risks; a mentor receives selected and omitted risks with source traces before a meeting [59]. HTCT turns tutor discourse into analytics that a coach selects and discusses with the tutor [17]. One carries the novice’s priorities into a conversation; the other makes teaching activity available for feedback. The studies follow different parts of the response: the entrepreneurship study observed real coaching meetings, whereas HTCT followed tutors’ subsequent practice over seven weeks.

Software feedback can also make a learner’s difficulty less visible to another person. In WriteAid (Stream 2), some recurring questions became less visible to the teacher when pupils sought software feedback, while the teacher gained time to circulate and new peer conversations occurred [106]. Assistance changed both access to information and the opportunities to respond. This suggests examining who no longer sees a question as well as who can now answer it.

The recipient’s response distinguishes a proposed connection from an observed change in practice. eaSEL can share a child’s reflective artifact with a parent and prompt a conversation, but its evaluation did not observe the whole parent–child exchange [130]. M-Powering Teachers, in contrast, provided mentors with feedback on their own talk; randomized access increased their uptake of students’ contributions and reduced their share of talk [34]. The latter establishes a change in teaching practice, while student learning remains a separate question.

The target of that response also matters. VizProg users diagnosed learners’ code and selected feedback in a classroom replay, but the feedback was not delivered to students [159]. QLens instead lets designers inspect prospective feedback; a recommended path derived from successful traces was judged unsuitable as direct instruction [155]. Recognizing a difficulty, improving one’s teaching, and deciding what will help someone else are different skills. The cases connect each to a specific representation and response. For shared support, the important sequence is what travels, who receives it, and what they do next.

Across these workflows, generation changes the material people start from, checking requires knowledge of both a result and its grounds, and handoffs require interpreting another person’s situation. These are concrete targets for development: composing with alternatives, judging a criterion, and responding to someone else’s reasoning. Their value as learning targets depends on the person’s goals and future work. They can accompany continued execution

skills rather than replace them. The next section asks how interface mechanisms make this work possible and what evaluations establish about performance and learning.

6 FINDINGS: SUPPORTING WORK AND LEARNING

Semantic scaffolding supports work by making task meaning available for people to inspect, judge, and change. The reviewed designs expose two choices relevant to learning: how a representation frames the task, and which operations it supplies or leaves for the person. We first map these design choices, then examine how people judge generated material, revise connected work, and learn through its use. Theory-lineage comparisons explain the rationales behind selected designs; evaluations establish their particular benefits and costs.

6.1 Representations make work available for inspection and action

A mechanism connects a representation of task meaning to a contribution the person can make. A goal expressed as an authored rule or an adjustable threshold makes different actions possible: stating an intention, changing a limit, or comparing its consequences. We therefore ask what task meaning is represented, who constructs it, and what the representation invites or enables a person to do. People, AI, or both can supply the structure. Figure 9 presents the 18 categories through these relationships, and Figure 10 compares their prevalence across technologies at the level of the 5 coded families; a paper can use several. Table 17 counts the papers in each category and how many of them concern or measure learning, and the category maps in Appendix C.1 lead to particular mechanisms and their source papers.

Prior design-space reviews similarly make interface choices visible: how abstract ideas take concrete form, or how an interface elicits a person's preferences [68, 101]. Scaffolding research adds the distinction between structuring a task and making important work available for the learner to confront [120]. The relationships below connect these design choices to the work people do and the outcomes studies assess.

Alternatives, relationships, and histories give people material to compare. Graphologue turns text into entities and links [63]; Crystalline gathers options, criteria, and evidence from the pages a developer reads into a comparison table whose cells lead back to their sources [85]. Sources, rationale, and state make different grounds inspectable. HaLLMark traces copied LLM text to its prompts but relies on writers to mark the passages they consider AI-influenced [55]; automatic provenance captures direct reuse more readily than inspiration.

Goals and editable material let people state intentions and change the work: Plume can lock text while it regenerates the explanation around it [84]. Recommendations, questions, and sequences instead give people a proposed action or a route through the task. OATutor selects practice and reveals hints [113], while WithYou shortens the pauses it inserts into speech as a learner's shadowing improves [161]; revealing more help and reducing help are different interventions.

A shared representation makes contributions and interpretations available to other people. Wikibench supports discussion that revises label definitions and inclusion rules [76]. The concerns combine: a shared comparison can help people understand alternatives, act on them, and discuss their reasons.

To use the taxonomy, start from a decision the person must make. Find the mechanism that supplies information or an action for it, then ask what response it requires and which outcome would show success. A provenance link, for example, gives a basis for checking a claim, but the person still judges whether the source supports the wording. The link shows traceability; accurate checking, or learning to judge sources, needs its own observation. Consider the same collection of passages: supplied labels give the person categories to interpret; asking the person to categorize passages elicits their distinctions; editable labels let them revise the scheme. Choosing among these designs begins with which operation matters to the person's work and development.

Representations shape participation

Read the task meaning and the person's contribution together.



Representations may be constructed by AI, people or both. Activities shown are design possibilities.

Marked categories combine operations with form/access, help timing or social conditions. These choices can vary across the other relations.

Fig. 9. Represented task meaning and possible human participation across eighteen mechanism categories. Each card pairs a representation with the contribution it enables or invites; the central schematic illustrates the interface operation. People, AI or both may construct the representation. Marked categories expose form, timing and social conditions that can vary across relations. These are design possibilities, not observations of use or learning. Original category memberships and all 110 distinctions remain in Appendix C.1. Explore the design atlas; search all 110 distinctions.

GenAI papers more often offer user control

GenAI versus traditional AI/ML, accounting for publication year and venue

2,453 papers with family coding and classified technology · 2019–2025

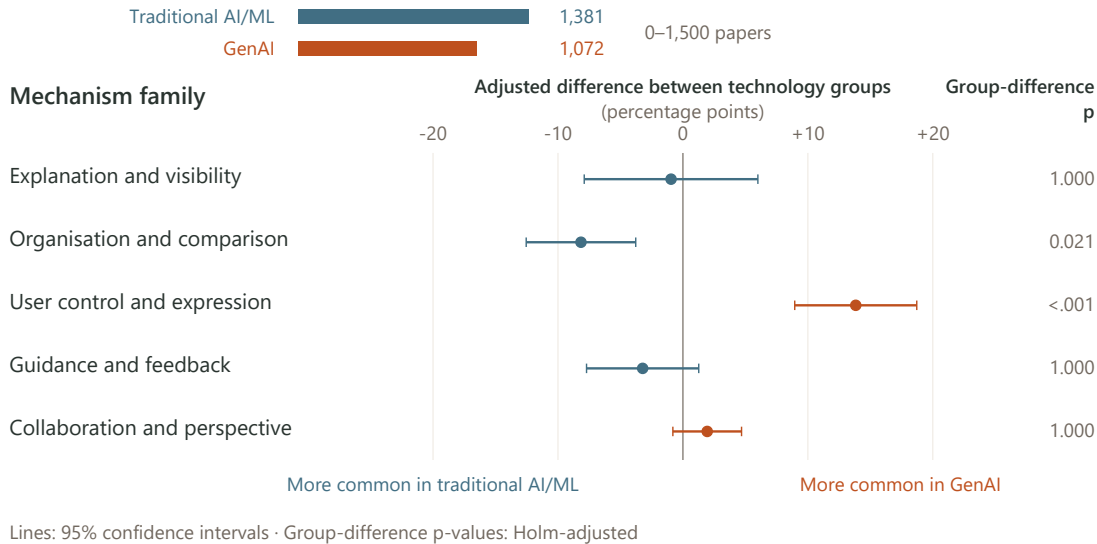


Fig. 10. Between-technology differences in mechanism families among 2,453 papers with usable family coding and classified technology, 2019–2025. Dots show GenAI minus traditional AI/ML prevalence, adjusted for year and venue; lines show pointwise 95% confidence intervals. The p -values test these group differences, with Holm correction across 51 contrasts, rather than change over time. User control retains support in the 2023–2025 and partial-2026 sensitivity windows. Of 3,424 family-coded papers, 544 lack classified technology and 427 classified papers fall outside 2019–2025.

6.2 Generated candidates still need criteria and grounds for judgment

More opportunities to choose generated material make the basis of a choice consequential. The quantitative patterns locate this design problem; the cases show how an interface’s alternatives and criteria shape what a person can examine and improve.

Technology and annual patterns. GenAI systems more often expose generated candidates. GenAI papers more often describe user control and expression (+13.8 points after adjusting for year and venue; 95% CI 8.9 to 18.7; Appendix D.2). In the 2,532 papers with facet codes, two facets in the control-and-expression family show a reliable gap, and both put generated candidates before the person: GenAI systems more often generate alternatives to explore (+4.8; 95% CI 3.0 to 6.6) and complete or continue the person’s own content (+3.0; 1.4 to 4.6; Figure 9). The separate retained-role analysis shows a positive difference for steering the loop (+3.4; 1.2 to 5.6) and a negative difference for improving the system (-5.9; -7.7 to -4.0; Figure 8). Other controls have different estimates: editing machine output (+1.8; -1.4 to 5.1), direct manipulation of a semantic object (+0.9), and specifying criteria and rubrics (-2.8; -4.9 to -0.8). Explicit criteria and rubrics are coded in 4.4% of GenAI papers. The generated alternatives alone do not establish how people judge them. Section 5.2 and Figure 6.2 compare designs that supply them.

GenAI systems rank items less often. Organization and comparison is less common in GenAI papers (Figure 10). Its largest facet gap is ranked or prioritized surfacing, which orders items so the important ones come first (3.2% of

Collaboration and perspective increase across years

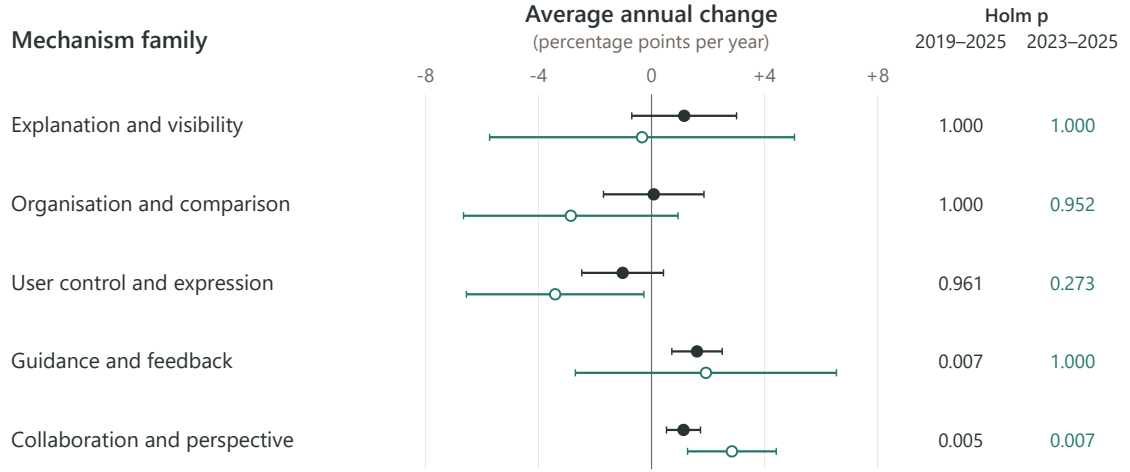
Annual change in coded prevalence, with a fixed mix of AI technologies

Adjusted for venue · 39 venues · Partial 2026 excluded

● 2019–2025 · 2,453 papers

○ 2023–2025 · 1,391 papers

Mechanism family



Lines: 95% confidence intervals · Holm correction across ten temporal tests per window

Fig. 11. Average annual change in coded mechanism-family prevalence: 2019–2025 (2,453 papers) and the 2023–2025 sensitivity (1,391). Dots show percentage points per year; lines show pointwise 95% venue-clustered confidence intervals. The technology mix is held fixed at 43.75% GenAI in both windows. Positive values indicate increasing prevalence. These are temporal slopes, not the overall technology-group differences in Figure 10. Appendix D.3 gives the methods, changing-gap estimates and all ten tests used for Holm correction within each window.

GenAI and 11.2% of traditional AI/ML papers; -6.5; 95% CI -8.3 to -4.7), and no other facet in the family differs by more than 2.9 points. Appendix D gives the annual family profiles (Figure 57).

Collaboration and perspective became more common over time. Holding the technology mix fixed at the original analysis's reference composition (43.75% GenAI), its average annual change was +1.13 percentage points (95% CI [0.53, 1.73], Holm $p = 0.005$). The 2023–2025 sensitivity also showed an increase (+2.84 points per year, CI [1.27, 4.41], Holm $p = 0.007$). Guidance and feedback increased in the primary window (+1.61 points per year, CI [0.72, 2.50], Holm $p = 0.007$), but its recent interval included zero ([-2.69, 6.54]). None of the five tests of change in the GenAI–traditional AI/ML difference survived Holm correction in either window. These temporal estimates (Figure 11) answer a different question from the overall technology-group differences in Figure 10; Appendix D.3 reports both temporal estimands for every family.

People may need to judge both an alternative and its criterion.

When a system generates the criteria as well as the alternatives, people must judge both. In Luminate, selecting an axis rearranges the responses, whereas adding a dimension revises the responses themselves [137], so a writer judges the dimensions as well as the ideas. VIVID generates expected answers and rubrics along with candidate dialogues, and instructors inspect and edit them [30]. Design-rationale research already links questions, options, and criteria [94], and conventional tools sort alternatives into categories that designers choose. Generated criteria raise a narrower

question: can people revise a criterion while holding the alternatives fixed, or does the revision also replace what they are judging?

Generated criteria can improve comparison. Selenite gives readers generated overviews of criteria and labels the relevant passages; in a twelve-person paired comparison with unaided reading, participants produced more valid criteria with the overview (12.93 versus 6.79), with higher precision and recall [86]. A criterion can also be easy to act on and still miss the quality it stands for. RECAST lets people revise text against a toxicity model; revisions lowered the model's toxicity scores without reliably improving human ratings of toxicity [152]. When both the model's response and human judgment matter, the evaluation has to measure both.

The supplied account can also direct attention away from what it omits. In a study with eighteen clinicians, automated suggestions shortened median label-finding time from six to three seconds. With prepopulated suggestions, clinicians annotated a smaller share of nontrivial spans the system omitted (medians of 58% versus 74% without suggestions) [83]. The aid helped one operation while changing attention to material outside its suggestions. Checking the supplied items and searching for missing ones are different contributions to the task.

A label can also leave the question itself ambiguous. In a prior sensemaking study (Stream 2), an unexplained "trust" lens led some researchers to consider a person's trustworthiness and another to consider that person's trust in sensor data [37]. The same displayed criterion can therefore invite different judgments; understanding its use requires following what people take it to mean. Across these examples, making a task inspectable also presents an account of what matters in it. The design choice is which criteria, relationships, and omissions the person can question; the learning target is the particular judgment needed to do so.

6.3 Iteration requires revising decisions and their consequences

Judgment continues as work changes. A revised criterion can call for new alternatives, and a corrected assumption can alter later results. The cases show why support for iteration depends on the links between operations as well as the individual controls.

Mechanisms can also be grouped by the activity they support: *specify intent*, *represent possibilities*, *evaluate alternatives*, and *act and revise* (Figure 12). Appendix C.1 maps all 110 facet codes to these activities; a code can serve several. Annotations, histories, shared views, and handoffs connect the activities across time and people. The functions describe dependencies, not a fixed order: comparing needs alternatives and criteria, and a comparison may prompt a change to either.

Different mechanisms can serve the same function. Intent elicitation, constraints, and shared criteria frame a task. Structured alternatives and source links make possibilities inspectable; comparisons, explanations, and reflection support judgment; selection, editing, and changed constraints enable action. One mechanism can serve two functions: a comparison exposes options and supports choice, and a constraint can frame a task or redirect it. Guidance can enter at any point, and histories keep context across transitions.

Luminate's dimensions organize alternatives for exploration and evaluation, with return paths to earlier prompt-specific spaces [137]. AI Chains instead feeds one step's output into the next step's input, so people can inspect, edit, and re-run intermediate steps [154]. The order in which a system computes need not dictate the order in which people inspect or repair its work.

An ordered workflow can still require return paths. CollabCoder opens shared comparison after independent coding, then supports discussion, decisions, and grouping while letting people restore their original codes [46]. Crystalline instead builds a route back into the material: a reminder marks related evidence on pages the developer

From task intent to action

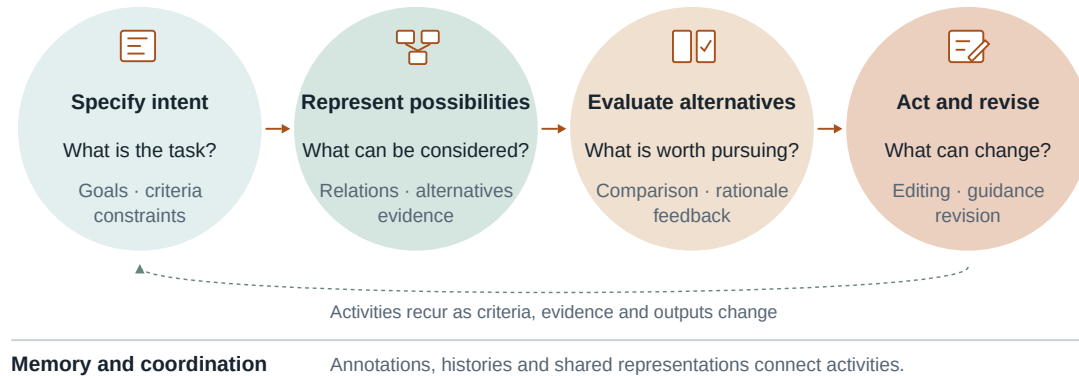


Fig. 12. **From task intent to action.** Four overlapping functions offer an activity-oriented view of the mechanism taxonomy. Memory and coordination connect them across time and people. Arrows show possible dependencies and return paths, not measured transition frequencies. The accompanying source examples illustrate these relationships [46, 85, 137, 154].

has visited but not yet attended to, and three of twelve participants reported using it to guide their research [85]. CollabCoder sets an order for the work; Crystalline returns people to an earlier step when their comparison may have a gap. Transitions therefore deserve the same design care as individual mechanisms: keep the criterion that shaped an alternative, the evidence behind a draft, and a route back to an earlier decision, and test whether people understand a transition, change it, and recover its context.

Scholastic shows why the effect of a control also needs to be understandable. Researchers' codes constrain keyword clusters, while linked document maps, readers, and code views support their interpretation. In a formative study with two researchers, the participant who saw a model update expected documents with the same code to cluster together. They instead remained spread across clusters: the constraint bound keywords, not document membership [54]. The mismatch concerns how the control's effect was understood: it constrained keywords while the researcher expected it to reorganize documents. Exposing a model update alone does not make that relationship clear.

What a revision preserves.

An editable step does not guarantee that people can correct it and keep the later work they still want. Across the cases, the design question is what a human action authorizes the system to reuse or treat as accepted. Table 1 distinguishes saving an old version, preserving valid edits during regeneration, and confirming a result.

Keeping an old version does not carry its edits forward. Kazemitabaar et al.'s data-analysis systems saved earlier versions in branches, but regeneration ignored later manual edits [66]. Participants described avoiding revisions and expecting to redo or reread work; one wanted invalidated parts highlighted instead of regenerated, and another valued the interconnected updates. Recovering an old version differs from preserving the still-valid parts of that work in a new one.

LineageD's pilot showed the cost of predicting a whole lineage at once: correcting one pairing invalidated the rest of the path. The redesign predicts one level at a time, but moving to the next level implicitly confirms pairings the person has not individually confirmed [53]. Plume keeps locked or manually edited text and uses it as context for later generation [84], so a lock preserves wording without limiting its influence on other text or vouching for its correctness.

Table 1. **Revision and downstream effects.** What three interfaces preserve, update, or treat as confirmed.

Design choice	Comparison and finding	Conditions and contrary evidence	Design implication
What survives regeneration?	In task decomposition, editing an assumption, plan or code component can regenerate later work while omitting manual edits [66].	The earlier version remains recoverable in a branch; preservation within the revised result is a different property.	Make the fate of earlier edits inspectable when an upstream decision changes.
What does locking text preserve?	Plume retains locked or manually edited dashboard text while using it to inform further generation [84].	An unchanged passage can still influence other text; locking does not isolate its downstream effects.	Distinguish preserving wording from restricting its influence on new content.
What counts as confirmation?	LineageD permits corrections before predicting the next level; advancing implicitly confirms remaining pairings [53].	Explicit and implicit confirmations share the green marking, so the display does not distinguish how a pairing was accepted.	Preserve whether an item was checked explicitly when that distinction matters to later work.

Relating a local step to the whole process is hard in itself: in AI Chains, four of twenty participants asked how a local change would affect the final result [154].

Interfaces promise to keep different things. Automatic propagation can itself help. CoPrompt participants valued automatic coordination, and eight used history and difference views to resolve conflicts or restore versions [40]. Pail’s code differences, summaries, and design panel supported inspection, but participants reported difficulty following cascades across them, less so when they paused to read the changes [157]. Selecting part of an artifact does not by itself fix what stays unchanged: FusAI leaves unselected image regions untouched, whereas Textoshop can change text beyond the selected passage [97, 116]. Blace keeps earlier states for rollback, and DesignManager records the process behind an asset [88, 156]; both support recovery and inspection of past work, while consistency with a revised goal still requires checking the downstream artifacts. A design should name the property it protects and the change that could violate it, and an evaluation should test whether people anticipate that change, detect a violation, and recover the work they want to keep.

Edits can change the specification, the model, or the output. A correction can change what the person asked for, how the system understood it, or what it produced. FusAI lets designers compose and revise a visual specification [116], and Falx synthesizes candidate charts from examples, which participants described checking against input values they knew [144]. CoAutoML lets novices inspect and edit the task, inputs, target, and evaluation assumptions it proposes before they confirm a machine-learning task [136]. Plume’s locks retain part of what the system produced [84]. Outside generative chat interfaces, corrections can also update the model: SmartEye uses a corrected crop as a positive training example for later recommendations [92], and DeepSI fine-tunes BERT from spatial feedback and recomputes its document map, though its authors note limited visibility into the model’s internal state [13]. The design question is what a correction changes and which later operation uses it, not whether every control comes before generation.

6.4 What AI supplies changes what people practice

Support can supply a representation, ask people to construct one, or use it as the basis for another activity. The reviewed comparisons show why the operation matters: receiving material can enable useful practice, while an extra requirement

Table 2. Selected AI-involved comparisons of explanation and practice designs.

Comparison	Assessed result	Conditions
Predicted-misconception explanations [21]	Better immediate exercise-prescription decisions after AI removal than unilateral explanations or no AI	Advantage over random foils unresolved; no delayed test
Required reflection after AI feedback [7]	2.0 versus 3.4 problems completed in eight minutes; transfer and code-evaluation advantages unestablished	Fixed time budget; fewer attempts do not establish learning harm
QuizBot versus flashcards [123]	Higher recognition (87.2% versus 65.8%), with an advantage at two months	Equal repetitions; 46 versus 14.3 minutes of practice; dialogue bundled with other support
Adaptive versus generic video recommendations [77]	Small algebra-ability advantage during normal schooling	Low optional uptake; policy effect; closure-period advantage unresolved

to reason or reflect can change both practice and its cost. Which arrangement supports learning depends on the particular capability being developed.

Learners need not build the material they learn with. VocabEncounter supplies phrases in context and reports better vocabulary two days later than its word-only condition [5]. Rehearsal supplies conversational alternatives, practice, and feedback; its participants used more cooperative strategies in a later unaided conversation than participants given the same static materials without interactive practice [129]. In a study of AI assistance for incidental learning, explanations without a recommended answer improved immediate concept learning over minimal feedback [44]. Supplied material also leaves room for different uses: CoNotate participants reported both running its suggested queries and borrowing their vocabulary to write new ones [111]. The design question is which operation the supplied material helps people perform, and whether the study tests that ability.

Explanation content, required reflection, dialogue, and optional recommendations change different parts of practice. Table 2 links four AI-involved comparisons to their assessed outcomes.

The practice cost of reflection does not explain every null learning result. In Xiaomai’s advanced-mathematics study, requiring error reflection produced neither a detected learning advantage nor a detected difference in attempts [39]. Together with the fixed-time comparison above, this makes the content and uptake of reflection worth examining alongside the practice it displaces.

Explanation design also concerns sequence. Incremental XAI supplies factors progressively during model-prediction practice, then tests prediction on unseen cases without the explanation display [16]. This separates the presentation of an explanation from what a person can do once it is removed.

Prior learning research separates creating an explanation from using it. In a randomized study of 120 learners, everyone wrote subgoal labels; those who then practiced with their own labels scored 21.6 of 25 on an immediate test, against 17.1 with expert labels and 18.8 with none. Instructional materials were removed for the test, while the programming environment remained available [95]. This comparison sharpens ExPeerience’s question (Section 5.2): which part of the task should the person learn to perform? Learning research has long stressed what support asks learners to do [28, 120]; the interface choice is what to generate and what to leave for the learner.

CoWriting Kazakh tests a related division of work: children either try writing a word in the new Latin script before seeing the robot’s correct spelling, or write it in Cyrillic and let the robot convert it. Both conditions show the robot’s model spelling. Across 67 children, letter knowledge increased during the single session, but the study found no overall

advantage for either conversion condition [125]. The design question is which operation the learner needs to practice and which feedback makes that practice useful.

NoteCopilot makes the amount of supplied structure a concrete comparison. In its thirty-person, within-person study, intermediate assistance yielded higher immediate understanding scores than either minimal or automated assistance when learners first answered without their notes [27]. The packages changed both the supplied material and the work of organizing it. This supports examining the division of work, without identifying engagement as the cause or establishing a delayed benefit.

Writing feedback can describe a revision or judge its quality. ArgRewrite classifies the purposes of a writer's earlier revisions without judging their quality, and asks writers to agree or disagree with those labels before and after revising again [1]. ArgueTutor pairs claim and premise labels with recommendations based on their balance [143], and LegalWriter adds advice on argument components while leaving the semantic fit between claim and conclusion to the writer [148]; both improved the quality of text written with them over their comparison interfaces. A description of a revision and advice about quality make different assertions, and one interface can offer both. A learning claim needs a test of the judgment or revision skill the feedback is meant to build.

6.5 Prior knowledge connects design choices to testable accounts

Prior knowledge can motivate a design, specify an activity, or inform an assessment. The same principle can therefore support different interface choices, and a concrete implementation can still leave its proposed learning benefit open. The lineage comparisons trace these connections: which prior claim informs the feature, what the person does with it, and which comparison tests the resulting account.

The source-use map (Figure 14) links 1,697 matched lineage cards to 43,733 source-paper pairs in the current analysis. A source recorded in two papers contributes two pairs. The records distinguish concrete uses in a design or measure, conceptual influences, and citation-only mentions. These roles describe how prior work enters a paper; they are not a ranking of theoretical quality. Papers without matched lineage evidence remain outside this comparison.

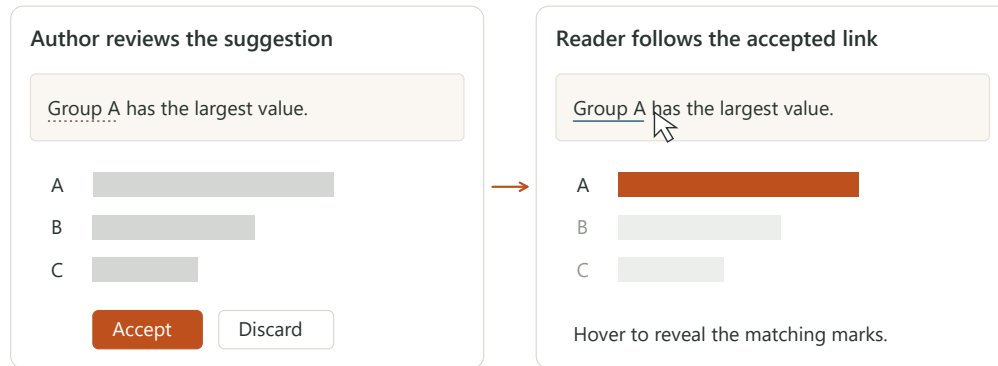
A rationale can leave the design choice open. DataDive draws on statistical thinking and critical reading to motivate contextualizing numerical claims. Its three design goals come from a seven-person formative study; the candidate contexts and ranking criteria are author-specified choices [69]. This establishes why context matters while leaving which generated questions help readers reason as an empirical design question. Its 21-person comparison found no detected advantage in data or data-interpretation recall over ordinary search; readers recalled more facts from outside the text in the baseline condition. The rationale, the interaction, and the assessed consequence support different parts of the account.

Making correspondence inspectable. Kori turns the concern about split attention into a visible link between prose and chart marks [78]. The system proposes links, and authors preview, accept, discard, or draw them by hand; once a link is accepted, hovering over its phrase highlights the chart marks it refers to (Figure 13). The author checks each correspondence, and the reader uses it to interpret the data.

Operationalizing a theory is not testing learning. QuizBot cites ICAP [28] as its learning rationale, whereas the Khanmigo methodology turns ICAP into an LLM-judged indicator of cognitive engagement [123, 140]. The studies assess different capabilities: QuizBot tests recognition two months later, while Khanmigo tracks correctness on the next attempted item. A direct test in diagnostic reasoning makes the distinction sharper: problematizing elicited more constructive responses than structuring, yet no condition advantage on diagnostic strategies was detected in subsequent cases after the pedagogical module was removed [50]. Both groups received active scaffolds, and the tests occurred

Kori makes text–chart links inspectable

AI proposes a connection. The author decides whether to keep it.



The author checks the correspondence; the reader uses it to navigate the data.

Fig. 13. **From an AI suggestion to an interactive reference in Kori.** The author reviews a proposed text–chart connection. After acceptance, hovering over the linked phrase highlights the matching marks by fading the others [78]. Schematic redrawn from the paper’s interface description, with illustrative data.

in the same session. A theory can identify a promising activity; observing more of it still leaves its contribution to capability to be established.

Prior knowledge can shape the evaluation as well as the design. Sibyl adopts Doshi-Velez and Kim’s application-grounded evaluation [35, 166], testing explanations with screeners in child-welfare work. Screeners often rated case-specific explanations helpful and found other forms less useful during those cases, and their feedback exposed problems of terminology and presentation.

A theory’s evidence travels with its comparison. Codigen cites cognitive mapping to motivate active learning and supplies generated programming material on an editable whiteboard [38]. The cited medical study compares maps with notes organized around the same five problem-solving elements, with the assigned support available during the diagnostic case [146]. It does not compare self-authored with generated material. Codigen’s novice pre/post quiz gains likewise do not isolate the whiteboard’s contribution. The connection suggests a design, while testing its learning rationale would require comparing how learners construct and use the material.

The comparison can change when a whole workflow is implemented. Dow et al. compared making three prototypes before critique with receiving critique after each one, holding prototype count, feedback, and time equal [36]. VisiBlends cites parallel prototyping within a rationale for process scaffolding, then supports generation, evaluation, and iteration of visual blends [29]. Its thirteen-person crossover study found more successful blends with the tool, without establishing an advantage in the next no-tool block. This tests a bundled production workflow; it does not isolate the earlier study’s critique order. The source suggests a design choice whose contribution still needs a comparison within the new workflow.

These cases locate different gaps in the connection from theory to evidence. Kori makes a design response to split attention inspectable; the ICAP comparison separates a theory-motivated design, an engagement indicator, and evidence

Table 3. Assessed outcomes in four AI-involved comparisons.

Comparison	Assessed result	Conditions
CodeVoyager versus same-model chat [70]	Higher immediate closed-book comprehension (8.06 versus 5.44/20); no detected assisted-completion or accuracy advantage	16 developers; interface package; no delayed test
FeedbackWriter versus usual TA feedback [87]	Better revisions ($d = 0.5$); knowledge-test advantage unestablished	Randomized 354-student trial; optional posttest; test-time assistance unclear
Rehearsal added to static training [129]	Better immediate conflict-response behavior; recall and recognition advantages unestablished	Behavior and recall assess different capabilities
CodeTailor versus generated solutions [57]	Better immediate unsupported application of supported elements; total posttest advantage unestablished	Selected 14 of 18 participants; tested elements depend on practice difficulty

of later capability. Codigen and VisiBlends carry a prior idea into a different activity or a larger package of support. These connections make the next empirical question concrete: which design choice enables the intended activity, and whether that activity develops the intended capability.

6.6 Studies establish different kinds of success

The reviewed studies report three kinds of outcomes: process, or what people actually did; artifacts, or the work produced with support; and learning, or what a person could do on a subsequent attempt. Required reflection left fewer attempts within a fixed practice interval, a process finding rather than proof of learning harm [7]. BELVEDERE's map and prose results below concern artifacts; the capability comparisons that follow test specific activities under specified assistance. Each result retains its own comparison and timing. Rating a finished artifact later still measures that artifact, not what its author has learned.

Product and capability measures answer different questions. In BELVEDERE, the maps showed more explicit links between evidence and hypotheses, but no difference in final prose conclusions was detected between conditions, and students could still consult their maps while writing [138]. Both outcomes describe products made with support; neither tests what students can do without the maps.

A benefit depends on the comparison and the situation. Cognitive forcing asks people to engage more deliberately with AI advice. Among 199 participants, fully correct decisions averaged 35% with simple explanations and 33% with forcing, a nonsignificant difference; on items with incorrect AI advice, forcing raised correctness from 3% to 9%, while the no-AI condition reached 18% [20]. Adaptive help raises the same issue: predicting whether a person will answer correctly differs from showing that advice based on that prediction helps them [90, 135]. Appendix A.7 gives the learning and duration profiles of the reviewed studies.

Assistance also defines the capability being tested: independence from a helper can coexist with continued use of a device. We return to that design choice in Section 7.2.

Performance and subsequent capability. An interface can improve one assessed outcome while leaving another unresolved. Table 3 distinguishes comprehension, revision quality, enacted behavior, and application of supported skills.

These comparisons apply Norman's distinction between performance with an artifact and the capability attributed to the person [107]. DBox's decomposition and feedback also improved performance on an immediate similar problem solved without assistance, although effort rose [91].

Assessment conditions specify the capability demonstrated. Among the 396 AI-involved sources in the outcome appraisal, 223 reported a separate capability assessment, 31 assessed capability in a later session, 71 used new material or a changed task, and 67 reduced or removed a named training aid. These overlapping features describe the selected appraisal set (Appendix A.8).

An assessment’s task, timing, and available support each change what it shows: an immediate new problem can test application without testing retention, and a later test can still allow help. WriteAid tested pupils’ writing two to three weeks later without AI or teacher support, but on the same topic and without an untreated comparison or a pretreatment skill measure [106].

The promised use determines the informative test. Immediate performance, later retention, and application to new material need not move together. Two classroom experiments with structured LLM reflection found no significant advantage on exams two weeks later [75]. Preference is not capability either: across three AI Margin Notes studies of reader interfaces, note selection, and human and AI note-taking, comprehension two hours after reading did not differ significantly, even where preferences or task demands did [64].

An informative evaluation therefore starts from the design’s intended use. An explanation meant to prepare a new decision needs a decision task; feedback meant to change the next attempt needs that attempt observed; a practice partner meant to build interpersonal skill needs evidence of enacted behavior, while recall answers a different question. Effectiveness reviews already show that scaffolding can improve cognitive outcomes [11]; the comparisons here show which alternative and which outcome a designer can use as evidence.

7 DISCUSSION

People should be able to use increasingly capable tools while continuing to develop understanding, skill, and judgment. What they need to learn will change with their work, goals, and available resources. Our review makes this design problem concrete: interfaces supply different material, leave different activities to people, and create different occasions for practice. We argue that semantic scaffolding should make those occasions meaningful, adapt them as people and tasks change, and connect claims about learning to the activity and evidence that support them.

7.1 Make developing capability a design objective

The workflow findings show why deciding what to automate also requires deciding what people should have opportunities to learn. CorpusStudio gives writers examples to interpret before composing; ABScribe gives them passages to compare and revise [33, 121]. Both support writing, but they place different demands on the writer. Norman’s distinction between system accomplishment and the person’s activity helps explain this change [107]. As the starting material changes, the person may need to develop a different combination of composition, interpretation, and editorial judgment. Which combination matters depends on the work they want to do. A learning objective should therefore be chosen with the person, alongside the task the tool helps them complete.

Generation can create practice in that capability. ExPeerience supplies problem contexts, tables, and explanations while leaving SQL construction to the learner [163]. Rehearsal supplies a counterpart and feedback through which a person practices responding to conflict. Compared with participants given the same static training materials, Rehearsal participants used more cooperative strategies in an immediate subsequent unaided conversation [129]. These designs use generated material to make an activity available: constructing a solution or trying a response and considering its consequences. The location of assistance matters because generating a problem, a response, or feedback gives the person different work to do next.

Meaningful practice needs both a worthwhile operation and support for performing it. Reiser distinguishes organizing a task from drawing attention to problems that require the learner to reason [120]. A worked example can make an unfamiliar problem approachable; a comparison can expose a distinction; feedback can help someone reconsider a decision. The design task is to connect these resources to the understanding or skill being developed. Simply leaving more work to the person supplies no such connection. In CoWriting Kazakh, children either tried writing in the new script or let the robot convert an entry in the familiar script; both conditions showed the robot's model spelling. Among 67 children, the conditions produced no detected overall difference in letter learning during the single session [125]. The target operation, its difficulty, and the feedback available together define the learning opportunity.

This gives the mechanism taxonomy a purpose beyond choosing an interface feature. Start with an activity through which the person could develop: explaining a relationship, comparing possible solutions, diagnosing a problem, or revising a judgment. Then choose the representations and actions that make that activity possible within the work. A generated explanation may support reasoning about an unfamiliar case; editable alternatives may support exploring the consequences of a choice. Semantic scaffolding connects those forms of support to a deliberate opportunity for the person to think and act.

7.2 Adapt scaffolds to changing work and developing people

An opportunity that is useful now may need to change as a person develops. Expertise-reversal research shows why guidance useful to a novice can become redundant as knowledge grows [65]. Work also changes within and across tasks: a new goal, audience, or source can make an earlier decision worth reconsidering. Scaffolds therefore need to respond to two forms of change. They should help people address the difficulty in the current task and support the capability they are developing across attempts. Successful completion alone gives an incomplete basis for deciding which change is needed.

Continuing access to a resource can be part of that development. Pea distinguishes developmental scaffolding with fading from enduring distributed intelligence [115]. The distinction directs attention to what responsibility changes and what resources remain useful. In vocational prompting, completing a task with a device counted differently from asking an immediate helper for assistance [25]. The person could develop a specified form of independence while retaining the device. A learning trajectory might similarly move from interpreting worked examples toward constructing and explaining a solution, with references available throughout. When the aim is to transfer that responsibility, its corresponding assistance can diminish while other resources remain.

The revision findings show what continuity requires from a representation. Plume can preserve selected wording during regeneration, while the reviewed data-analysis systems can retain a recoverable earlier version yet omit its manual edits from regenerated work [66, 84]. These differences matter when people return to a decision: they need to recover their contribution, understand what changed, and decide which parts still fit. Malleable interfaces and persistent organizing rules offer ways to reshape a working representation [67, 100, 102]. Maru's persistence can also carry a rule from an earlier subtask into a setting where it no longer fits [67]. Support for development should let people revisit their reasoning, preserve useful work, and revise or retire an organizing rule when its purpose changes.

Adaptation also involves the people who help someone learn. The workflow findings trace support through novice-mentor and tutor-coach exchanges [17, 59]. A representation can make a difficulty available to someone able to respond. It needs to carry enough context for that person to understand the learner's priorities. Designing its evolution therefore includes deciding what travels between people and what each can question. Together, these cases motivate support that

Table 4. Proposed evaluation guide by collection time and evidence source.

Evidence source	During the task	After the task
Observed or externally assessed	Actions, revisions, errors, task time	Artifact quality ratings; scored understanding, retention, transfer
Reported by participants	In-the-moment effort, confidence, agency	Retrospective effort, usefulness, perceived learning, self-efficacy

preserves enough context for people to reconsider earlier work while making room for greater capability, changed goals, and new contributions from others.

7.3 Connect learning claims to activity and evidence

The proposed design commitment makes three consequences relevant: what people do with support, the work they produce, and the capability they develop. The Findings show why these need distinct evidence. Among sixteen developers, CodeVoyager improved immediate closed-book comprehension relative to same-model chat, with no detected assisted-completion or accuracy advantage. FeedbackWriter’s randomized 354-student comparison with usual TA feedback improved revisions; its optional knowledge test left a learning advantage unestablished, with test-time assistance unclear [70, 87]. An evaluation can use these different outcomes to understand what a design changes and whether it serves the intended learning objective.

Theory can explain the connection to look for. Kori translates a concern about split attention into inspectable correspondences between text and chart marks [78]. That connection identifies a feature to examine and an interpretive activity it is intended to support. The diagnostic-reasoning comparison follows the account further: problematizing elicited more constructive responses than structuring, while the two active-scaffold conditions showed no detected difference in diagnostic strategies on subsequent same-session cases after the pedagogical module was removed [50]. The activity observation and capability assessment locate different parts of the explanation. The theory lineage is useful when it makes these relations explicit: which construct guides the design, how the interface engages it, and what evidence would test the proposed consequence.

Process, artifact, and learning measures can then be chosen to examine that account. Process observations show how people use a representation, including the practice and checking it elicits. Artifact assessments show whether the work meets its purpose. Capability assessments examine the intended activity on another attempt, under specified conditions of task, timing, and available help. Participant accounts describe how people interpreted the task, what they valued, and what they found difficult. Combining them can explain why support was taken up, ignored, or used in an unexpected way.

We propose two further measurement choices: whether evidence is collected during or after the task, and whether it comes from observation or external assessment, or from participants’ reports. Table 4 presents these choices as a guide for planning evaluations.

Collection time and the subject of a measure answer different questions. A retrospective report of effort describes an earlier activity; a later quality rating still concerns the artifact. A learning claim needs an assessment of the intended capability, with the comparison and remaining resources specified. External ratings depend on scoring criteria, and a numerical self-rating remains a participant report. For each measure, the construct, instrument, rater, and assessment conditions explain what evidence it contributes. The proposed matrix organizes these choices without adding new classifications of the reviewed studies.

This combined account helps designers judge a learning opportunity in relation to useful work. Extra explanation or practice takes time; its value depends on what the person learns and what the task requires. An evaluation that follows activity, assesses its products, and examines subsequent capability can inform whether to retain a scaffold, change its form, or offer it at a different moment. It also gives future studies a specific explanation to extend or challenge.

8 RESEARCH OPPORTUNITIES

The review opens a research agenda around creating meaningful learning opportunities as work changes. Four questions concern the designs that make an activity possible, explanations of how that activity supports learning, the formation of judgment across people, and the place of learning in everyday tools. Together they connect interface development with an account of how people develop through use.

8.1 Design representations around meaningful learning activities

The design space offers ingredients for supporting comparison, explanation, diagnosis, construction, and revision. Research can investigate how to combine them around a capability a person wants to develop. For example, two worked solutions can make a relationship available for comparison; a partial representation can invite someone to supply the missing relation; a history can let them revisit why they changed their interpretation. Each arrangement connects task meaning to a different operation. The opportunity is to develop a repertoire of such connections, including the feedback and prior knowledge that make the operation productive.

The existing comparisons expose design choices within that repertoire. In the subgoal-label experiment, all learners created labels; those who then practiced with their own labels did better on an immediate test than those practicing with expert labels or no labels. Instructional materials were removed for the test, while the programming environment remained available [95]. The comparison concerns what people use after construction as well as what they construct. Likewise, a ready-made criterion can reveal a useful distinction, as in Selenite [86], while revising a criterion asks the person to decide what matters. The assistance dilemma makes the broader question explicit: which support enables progress, and which operation is worth leaving for the person to practice [73, 120]?

Research through design can explore these choices with people working on problems they care about. Prototypes could let a person move between an example, their own construction, feedback, and a revised attempt while retaining the relationships among them. Work with users can identify where a representation opens a useful question and where it makes that question harder to understand. Focused comparisons can then examine a particular choice, such as supplying an explanation before or after a first attempt. The contribution would be reusable designs with an account of the activity they support and the situations in which that activity is worth practicing.

8.2 Develop explanations of how scaffolds support learning

The theory lineage creates an opportunity to connect a construct, its implementation, and the evidence used to judge it. A principle can inspire several representations, and similar representations can engage different activities. Codigen's cognitive-mapping rationale illustrates the work needed between these steps. Its cited medical study compared maps with notes organized around the same five problem-solving elements, with the assigned support available during assessment [146]. Codigen instead lets learners arrange and edit generated programming material; its novice pre/post quiz gains came from an activity without an unexposed control [38]. A useful next account would explain which part of constructing, organizing, or using a representation supports the capability in each setting, and identify a comparison that tests that explanation.

Practice conditions belong in that account. QuizBot compared equal numbers of repetitions while allowing substantially different practice time. Required reflection after AI feedback instead consumed part of a fixed eight-minute interval, leaving fewer completed problems [7, 123]. In Xiaomai, reflection produced neither a detected learning advantage nor a detected difference in attempts [39]. Together these comparisons motivate examining what people do during the added activity, how they use its feedback, and what other practice it changes. The explanation must distinguish these possibilities to inform another design.

Such research can combine close observation with focused interventions and comparisons across tasks. Observation can reveal how a representation is interpreted; an intervention can test an activity the explanation identifies; later assessments can examine the intended capability under the resources available in future use. Equal-time and equal-repetition comparisons address different practical questions where the activity changes exposure. For optional support, the offer and its uptake also need separate accounts [77]. The proposed evaluation guide in Table 4 helps select complementary evidence. Across a program of studies, the aim is an explanation that accounts for both productive use and failure, and helps anticipate when a scaffold will need a different form.

8.3 Study how people form and share judgments in use

Semantic scaffolds also offer a way to study how interpretations develop through collective work. A shared label or comparison brings people's reasoning into contact, while leaving room for different meanings. The sensemaking study's "trust" lens referred to different relationships for different participants [37]; CollabCoder brings individual codes into shared discussion [46]. Distributed cognition provides a basis for following that reasoning across people and representations [52]. The research question is how an interpretation becomes useful, contested, or revised as people try to produce work together.

Handoffs make this process visible over time. Entrepreneurship coaching carries novices' selected and omitted risks into a mentor meeting; HTCT makes tutor discourse available to a coach who selects and discusses it [17, 59]. Studies can follow what a recipient infers, what the person represented corrects, and how their exchange changes the next attempt. WriteAid adds a different relationship: software feedback made some recurring questions less visible to the teacher while creating time to circulate and occasions for peer conversation [106]. A new route for assistance can therefore reorganize the opportunities for people to understand and respond to one another.

Studying these changes calls for observing ordinary work, including deadlines, disagreements, and the demands of producing an artifact. Traces can locate a revision or an abandoned alternative; observation and participants' accounts can explain its meaning. Follow whose interpretation enters the shared representation, who can challenge it, and what people carry into a later decision. Learning claims can then be examined for the people concerned, alongside changes in the group's practice and produced work. This would explain how scaffolds participate in the development of judgment within a social setting, including the conditions that make useful differences available for discussion.

8.4 Create learning opportunities within everyday tools

A tool for writing, programming, or analysis could help people develop capabilities through the work they already want to do. A consequential decision creates a reason to examine an explanation, compare alternatives, or try a revision. The incidental-learning study gives one bounded example: explanations without a recommended answer improved immediate concept learning relative to minimal feedback [44]. The broader opportunity is to make such moments available across ordinary use, where people also face immediate goals, limited time, and changing reasons to learn.

This requires designing a continuing relationship between the tool and the person's learning goals. A user might want to recognize misleading comparisons, diagnose a class of programming errors, or make stronger written arguments. A tool could help them notice relevant moments, retain earlier attempts and explanations, and return to an unresolved question when time permits. Malleable interfaces and persistent information architectures offer ways to change the invitation while keeping a useful working representation [67, 102]. People need to be able to choose, defer, and revise their goals as well as the support. Declining a practice opportunity during an urgent task is information about that situation; the tool would need further evidence to infer mastery.

Longitudinal design work can investigate how these opportunities accumulate and how support should evolve. Co-design can establish which capabilities people value and what resources their work requires. Field studies can then follow the opportunities encountered and taken up, interruptions, changes in work quality, and direct assessments of the intended capability on new cases. Assessments should retain resources that normally belong to the task and identify the particular training aid being varied. Following people as their goals, tasks, and expertise change would reveal when a scaffold helps them develop, when it needs revision, and which resources remain part of capable practice. That knowledge could make learning a sustained possibility within useful tools.

9 LIMITATIONS

Coverage. We searched 40 venues over 8 years, which favors published, English-language research [6, 80]. Technology comparisons leave out 645 papers whose AI technology could not be determined, and each analysis uses the papers coded for its fields (Table 9). The theories and empirical studies we compare against are a selection, not a census of every relevant tradition.

Reading and coding. Language models read each paper against a fixed codebook, one model per paper, and grouped the coded descriptions into categories. People have not yet checked the coding or the cross-case interpretations. Models can miss evidence shown only in figures and can share biases, and other analysts could draw the boundaries between closely related mechanisms differently.

Effects. Studies differ in tasks, participants, comparators, outcomes, and duration, and many compare whole systems. Most coded changes in how people work rest on the authors' own attribution rather than a measured contrast (75.8% of coded process shifts). The GenAI and traditional AI/ML comparisons are associations among published designs, not effects of the technology. The design space cannot rank mechanisms by effectiveness or estimate an overall learning benefit.

10 CONCLUSION

As AI changes work, people need opportunities to develop the understanding, skills, and judgment that let them pursue what matters to them. This review connects the structures AI interfaces supply with the activities people carry out and the consequences studies assess. The design space makes concrete choices available for creating meaningful practice within useful work: what to provide, what to invite people to do, and how to support the next attempt. Developing these opportunities requires attention to both the person's changing capability and the demands of the task. The research agenda is to understand how semantic scaffolds can evolve with that relationship, helping people learn through everyday tools and with one another.

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A EVIDENCE EXTRACTION AND CODING

This appendix holds what a reader needs to check or reuse the coding: the fields of each evidence card, how the closed and open fields were coded, the agreement checks, the data behind each analysis, how sources for the outcome comparisons were chosen, and how cited sources were matched.

A.1 Card Fields

Stream 1 mechanism cards (Table 5) describe the design, and theory-lineage cards (Table 6) describe its use of prior work. Each Stream 2 source has a knowledge-claim card (Table 7). Table 9 distinguishes the available evidence for each analysis. The sectioned layout follows Model Cards [103], and AI-assisted design-card generation is a precedent for turning source findings into cards [133].

Table 5. Stream 1 mechanism card: work and human activity first, then scaffold design and evaluation.

Card section	Fields recorded
Workflow changes (RQ1)	Process before and after support; mechanism credited; attribution basis; source of the baseline.
Human activity (RQ1)	Activity externalized to the tool or representation; activity retained by people; work still required.
Scaffold design (RQ2)	Author term; facet and value, with their merged codes; what makes each mechanism separate; supporting passage and page.
Learning and support (RQ2)	Learning measurement, direction, and evidence; reliance; changes in support and their grounds; observation horizon.
Measures and comparisons (RQ2)	Recorded outcome layer (process, artifact, learning); author claim, claim scope, contribution type, study design, participants, comparison, statistics, and claim-evidence alignment.
Conditions and limitations	Author-stated limitations and boundary conditions.
Recorded codes	Work domain, intent, expertise, capability gap, support properties, AI involvement, and technology era.
Source	Paper ID, DOI, venue, year, page count, the PDF read, the full-text decision, a map of relevant pages, and codebook notes.

Table 6. Stream 1 theory-lineage card: how the paper uses the sources it cites (RQ2).

Card section	Fields recorded
0. Identity and position	Link to the mechanism card; overall position toward prior work; position summary; human-facing rationale; lineage depth.
1. Documented sources (RQ2)	One entry per cited work: citation identity, relation to the focal paper, use, enactment, and supporting passage and page.
2. What was read	Introduction and related-work pages; design or evaluation pages checked for application of a cited idea.
3. Synthesis note	A short account of what the paper’s use of prior work shows.

A.2 Closed Fields

Closed fields take values from a fixed list. Intent records whether the support targets immediate performance or sensemaking, learning or skill development, or both. Measurement records what the evaluation assessed, so an intention

Table 7. Stream 2 card: one to eight field-level claims, linked to the research questions they inform.

Card section	Fields recorded
0. Identity and position	Card and work IDs; publication details; discovery routes; contribution types; position summary; overall uncertainty.
1. Page-grounded claims	One entry per claim: author wording and reviewer interpretation; question links; applicability; affected human work; outcome evidence; evidence basis; counterevidence and limits; supporting excerpts; uncertainty.
2. Evidence store	Verbatim excerpts with pages and locators; links recording which claim each excerpt supports, its role, and the reviewer paraphrase.

to support learning stays apart from a measure of it; the direction of a result and its observation horizon are recorded with the measure. Support properties are coded separately and can co-occur: structuring, problematizing, contingency, adaptation, fading, responsibility transfer, and distributed support. Contingency responds to the person's activity or need. Fading reduces support as competence grows; a fixed reduction on a schedule is not by itself fading. Responsibility transfer concerns who performs or regulates the activity, not whether the interface remains available. Technology is classified once per paper, from the AI in its system, an experimental condition, training, or evaluation, so a paper classified as generative AI can still contain mechanisms that involve no generation.

A.3 Open Fields

The 21 open fields hold 142,877 observations in the papers' own terms. Equivalent descriptions within a field were merged into 1,290 codes, and the codes were grouped into 223 categories and 92 families across 18 fields, with subcategories only where they add a distinction. Each grouping keeps links to its observations, and fields are never pooled. Codes that record missing information, such as an unspecified baseline or an unreported measure, stay apart from substantive categories. When one free-text value lists several limitations, its single code summarizes them rather than listing each. Appendix B defines the families and categories of the six fields the framework uses.

A.4 Agreement Checks

To test the open-code grouping, a separate model pass coded extracted observations with the existing codes hidden, and we compared the two groupings field by field (Table 8). An observation entered the comparison only when its identifier, source card, and field value matched exactly. Two codings need not use the same words, so we compared partitions: whether each pair of observations shares a code in each coding. The adjusted Rand index (ARI) corrects pair agreement for chance, and positive-pair F1 summarizes overlap among the pairs either coding groups together; neither treats one coding as correct.

Raw agreement is high because both codings keep most pairs apart: for facet, 19,083 of 19,503 pairs are apart in both and 32 are together in both. The blind pass used fewer codes than the existing coding in five fields and more in learning evidence. It reported medium or high uncertainty on most assignments in five fields (for example, 659 of 695 values) and on 7 of 24 in learning evidence. The learning-evidence row covers 24 of 48 existing observations; the other 24 share a repeated not-measured value and fell outside the comparison's selection rule.

A separate check asked only whether AI was involved: three fresh model sessions labeled 532 title-and-abstract records added in September and gave the same label to 529 (99.4%; $\kappa = .988$).

Table 8. Blind open-code comparison on exactly matched observations. Codes gives existing / blind code counts; Rand is raw pair agreement, and pair F1 compares the pairs grouped together. Unmatched counts existing observations outside the comparison. Papers overlap across fields, so the 1,571 matched observations are not 1,571 distinct papers.

Field	Obs. (papers)	Unmatched	Codes	Rand	ARI	Pair F1
Externalized to tool	110 (110)	18	64 / 46	96.2%	.167	.186
Facet	198 (48)	15	132 / 76	98.0%	.132	.142
Process before	125 (48)	15	79 / 34	95.8%	.445	.462
Process after	419 (160)	0	262 / 60	97.0%	.079	.088
Value	695 (144)	0	517 / 95	98.1%	.082	.087
Learning evidence	24 (24)	24	5 / 16	50.4%	.108	.217

Table 9. Data behind each analysis. Counts refer to the stated unit.

Data	Count and unit	Analysis
Stream 1 corpus	4,638 papers on 28 September 2026; 475 non-AI comparison papers outside the systematic counts	Corpus counts (Section 4)
Main figures	4,133 papers (4,162 records) with usable coding	Annual, technology, and mechanism figures
Closed fields	2,891 papers coded by 28 August 2026; 301 measure learning	Intent, learning-measurement, and process counts
Open fields	142,877 observations in 21 fields from the 3,295 papers coded by 28 August 2026, of which 2,847 remain in the corpus	Design space and framework values; counts of papers carrying a value
Complete mappings	2,315 papers with complete facet-value codes; 2,311 also with remaining-work codes	Mechanism-by-work matrix (Appendix C)
Theory lineage	1,697 PDF-matched lineage cards; 43,733 recorded source-paper pairs	Roles of prior work (Section 6.5)
Stream 2	512 sources; 2,867 claims	Prior findings and theory
Citation network	4,751 papers from both streams	Citation atlas and co-citation map (Section 6.5)

A.5 Validation Status of the Readings

For part of the evidence, the records of the original reading runs have not been fully validated, and we mark that evidence for provisional use. On 26 September 2026, such evidence appeared in 319 of the 1,076 learning papers and 1,252 of the 4,086 papers behind the main figures. The counts mark incomplete validation records, not errors, and the marked papers stay in the analyses.

A.6 Data Used for Each Analysis

Each analysis uses the papers coded for the fields it needs (Table 9).

All 2,847 open-coded papers have complete domain and retained-work codes; 532 lack complete facet-value codes. The closed-field population includes 44 papers outside the open-field profile and two with unresolved support-property coding.

A.7 Learning Measurement

Among the 2,891 closed-field papers, 301 (10.4%) measured learning. Two-thirds of those (199, 66.1%) measured it within a single session; 35 used repeated sessions, 38 followed people over weeks, and 25 measured learning in deployed use.

A.8 Selection of Sources for Outcome Comparison

The study comparisons in Section 6 draw on sources chosen for their outcome evidence. A learning-scope assessment of 4,714 records proposed 1,226 within scope. We appraised the 422 of these whose cards recorded that learning was measured (twelve more had been appraised earlier and were analyzed separately), plus 14 performance cases found among 53 descriptions that needed interpretation: 436 records, or 434 distinct PDFs. The 190 sources flagged as having a documented capability comparison entered the detailed synthesis.

A.9 Source Matching and Coverage

A.9.1 Source-use coding. Each theory-lineage record links one cited work to its role in one citing paper’s design, argument, or measure (Section 6.5), so a work cited by two papers yields two records.

A.9.2 Citation metadata. The bibliographic map joins the two streams by DOI or Semantic Scholar paper ID and counts a paper found in both once. It uses Semantic Scholar data retrieved on 9 September 2026 for 4,331 Stream 1 papers and 448 Stream 2 works (28 in both). Reference lists for 3,233 papers name 106,523 cited sources; two with conflicting identities are withheld. Of the remaining 106,521, 5,748 are cited by at least five review papers and 157 by at least 40 (Figure 15). A reference list shows that a paper cites a work; its lineage card shows how the paper uses it.

The co-citation map counts each citing paper once per source and keeps sources with at least five citers and pairs with at least three shared citers. It weights a pair by association strength: the pair’s shared citers times the number of papers, divided by the product of the two sources’ citer counts. Louvain clustering (seed 42) forms the neighbourhoods, and a maximum spanning forest supplies the drawn links; stem, branches, and leaves mark how widely a source is shared. Less-cited sources sit near their most shared neighbourhood without joining it. The interactive atlas supports zooming into a neighbourhood and inspecting each source.²

A.9.3 Matching and unavailable sources. Cited works without a DOI are matched by title and year. Title variants can split one work into several identities, which overcounts distinct works and undercounts shared references; descriptive labels and conflicting identities stay out of the shared-reference map. Papers without a reference list, and full texts we could not obtain or read, are outside the source-use analysis.

B THE FRAMEWORK’S CODEBOOK

The framework combines mechanism design with five dimensions of human work and evidence. Tables 10 and 11 define those five dimensions; Appendix C.1 gives the mechanism categories and examples. Counts use the 28 August 2026 coding populations in Appendix A.6; a paper may carry several values. Each group names the paper denominator for its percentages. Daggers identify absent or unspecified values, as defined in each row.

B.1 What follows for the person, the work and the evidence

²Interactive citation atlas: the interactive citation atlas .

Table 10. The three dimensions about the person: what the person still does, keeps, and hands to the tool, with the share and number of papers carrying each value. A dagger marks the absence of a value.

Dimension and value	Definition	Share	Papers
What the person still does (n=2,842 papers)			
The task the tool did not do	Work of the task itself that the system does not do: performing the substantive task, making sense of what is shown, and checking and repairing what comes back.	51.1%	1,451
New work the tool created	Obligations that exist only because the system exists: feeding and instructing it, and operating and coping with it.	29.9%	851
The whole task kept by the person	The person feeds the system and judges what comes back, across three or more stages, with no stage removed.	10.6%	300
Final authority kept by design	The final decision, the reliance decision, values-laden judgement and accountability stay with the person as a matter of design or law, not as a step in a workflow.	8.3%	237
Duty asserted, nothing specified†	The value says responsibility remains with the person without naming any obligation.	0.1%	3
What the person keeps (n=2,847 papers)			
Judgement and decision	Interpreting and inferring, deciding and authorising, setting direction, bringing knowledge.	48.0%	1,367
Doing and revising the work	Doing the work, repairing and revising, steering the loop, improving the system.	35.7%	1,016
Checking the output	Checking output for errors, calibrating reliance, judging fit to goal, verifying against outside evidence.	9.1%	260
Responsibility, social and self-directed work	Coordinating and deliberating with others, consent and disclosure, ethical judgement, communicating outward, holding accountability, care work; and learning, reflection and sustained attention to the system.	6.9%	197
Not stated†	The paper does not say what the person retains.	0.2%	7
What moves into the tool (n=2,326 papers)			
Computation and judgement	The tool predicts, simulates, fits models, calculates, optimises, estimates uncertainty, computes geometry; and attributes behaviour, classifies, detects errors, critiques, scores, infers a person's state, ranks and verifies.	30.5%	709
Representation and translation	The tool gives external form, converts between formats and languages, lays material out, summarises, verbalises, encodes data visually, coordinates views, compiles specifications; and finds groupings, patterns, similarities, aggregates and comparisons.	23.6%	549
Decision, guidance and action	The tool recommends, selects the next step or decides outright; supplies templates, procedures, decompositions, reference material, instruction and prompts; propagates judgments, does clerical work, operates environments, enforces constraints, fabricates; stands in for people, coordinates them, relays communication.	15.6%	364
Perception, search and memory	The tool senses, extracts, recognises, segments, transcribes and flags; finds and narrows material; records, stores, recalls and reminds; watches and tracks.	15.0%	349
Content generation	The tool generates new artifacts (alternatives, images, prose, code, 3D forms, audio) and responses (explanations, questions, answers, utterances, conversational replies, revisions of the person's material).	14.9%	346
Offloading that failed or was withheld	The offloading failed or was incomplete, or the tool deliberately stopped at a proposal and left the judgment with the person.	0.3%	8
Nothing offloaded, or not stated†	The value states that nothing moved to the tool, or names no operation at all.	0.0%	1

Table 11. The two dimensions about the work and the evidence: what happens to the work and how learning is evidenced, with the share and number of papers carrying each value. A dagger marks the absence of a value.

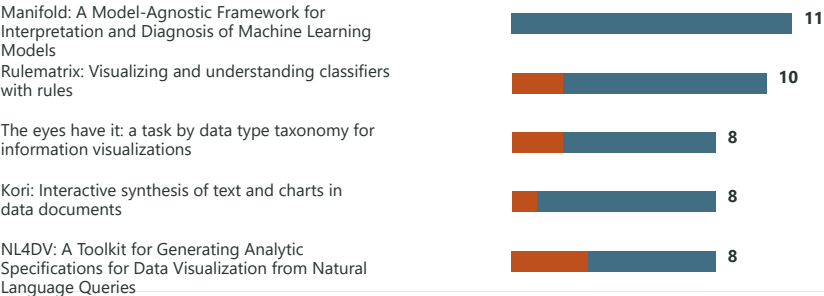
Dimension and value	Definition	Share	Papers
What happens to the work (n=2,643 papers)			
The person's work changes form	The person still works, differently: reading new representations, navigating and searching, steering and specifying at a higher level, reviewing and selecting instead of producing, checking and repairing, following a guided procedure, judging and reflecting on new grounds, coordinating through shared artifacts.	92.2%	2,437
A step becomes automatic	A step the person used to do is now done, or mostly done, by the system: automatic extraction or generation, exception-only attention, the same task made cheaper, automatic checking of the person's own work, effort freed for other work.	26.4%	697
Study condition or proposal, not practice	The value describes a study arm, protocol or manipulation, or a proposed or recommended change that was not observed in use.	16.1%	425
Work newly possible	The person can now do something they could not: work in a new setting or moment, at a larger scale or scope, or a task that was out of reach.	4.6%	121
New work created by the system	Labour that did not exist before the system: setting up, configuring and maintaining it, added steps or obligations, burden or unproductive activity it caused.	4.0%	105
Work unchanged	The practice did not change: the person keeps the core task and the system only informs it, under-uses or works around it, or nothing changed at all.	2.0%	54
How learning is evidenced (n=2,800 papers)			
A proxy measured instead	Capability stood in for by a proxy: subjective judgements about the system, usability instruments, qualitative accounts, attitudes or engagement; in-session task performance, behaviour or rated work products; or the quality of the system's own output, including the machine's learning.	37.6%	1,054
Learning measured	An objective capability measure was taken: immediately with support present, later or with support withdrawn, or with a null, mixed or adverse result, or compromised by confounding or deliberate suppression.	9.4%	263
Learning self-reported only	A learning or skill gain is asserted on self-report alone, or gestured at in remarks while the paper says learning itself was not evaluated.	4.6%	130
Learning not measured†	Nothing about the person's learning, transfer, retention or capability was measured: the value names no measure and no substitute, the authors say so themselves or defer it, there were no participants at all, or learning is the stated purpose yet never evaluated.	48.3%	1,353

Where prior work is used

1,642 distinct papers · 1,697 PDF-matched lineage cards

Most shared identifier-linked sources

Distinct papers with recorded use



Follow a source into later designs

Highest-ranked source with an unambiguous recorded year; every use is shown.

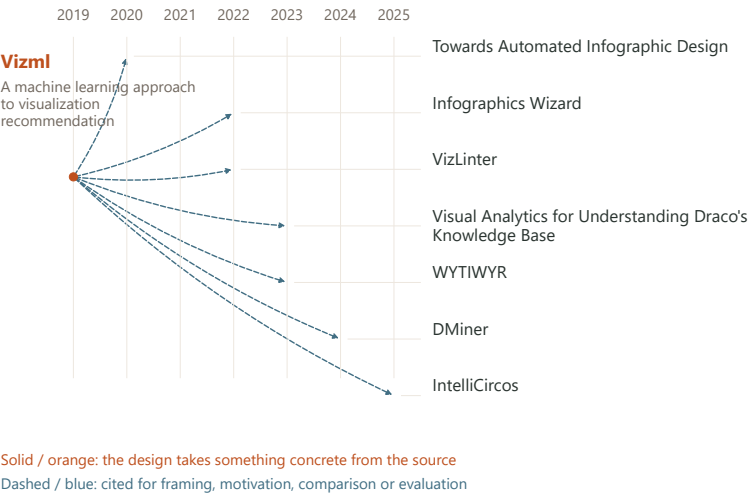
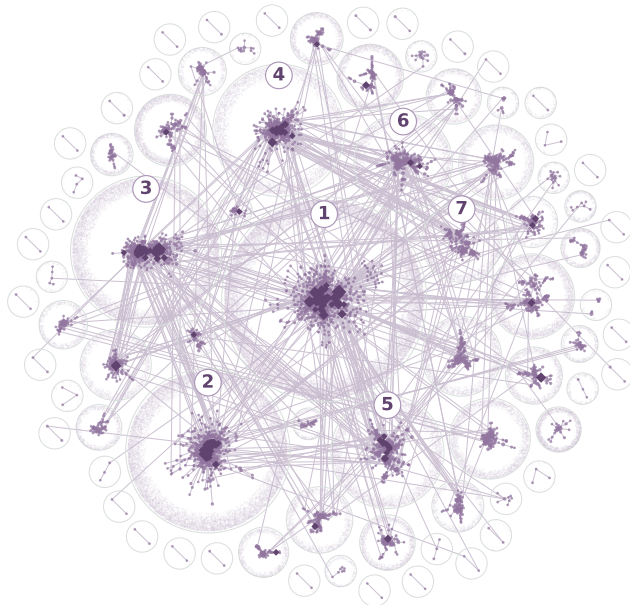


Fig. 14. **How papers use prior work.** Across 1,697 lineage cards for papers in the analysis, the upper panel ranks sources with a DOI or other identifier by the number of papers that use them. Orange marks papers whose design takes something concrete from the source (a mechanism, interface element, algorithm, representation or interaction); blue marks papers that cite it for framing, motivation, comparison or evaluation. Two language-model coders applied this one definition to every pair shown ($\kappa = 0.90$). Below, the top-ranked source with a known publication year connects to later designs, with solid arrows for the first kind of use and dashed arrows for the second. The lineage records link 43,733 source–paper pairs to passages and cards.

Shared sources across the two review streams

106,521 cited sources · 5,748 cited by five or more review papers



◆ Stem · 40+ citers · 157

● Branches · 5–39 · 5,591

Reference coverage: 3,233 of 4,751 distinct review papers · 9 September 2026

Groups follow co-citation; topic labels summarize prominent member titles.

7,369 sources without group connections remain available in the online source list.

Seven largest groups

- 1** Generative AI and human-AI interaction
789 shared sources
 - 2** Explainability, trust and AI decisions
567 shared sources
 - 3** Deep learning and model inspection
492 shared sources
 - 4** Visualization authoring and exploration
460 shared sources
 - 5** Responsible AI and research practice
322 shared sources
 - 6** Programming and data work with AI
247 shared sources
 - 7** Uncertainty and visual communication
165 shared sources
- Leaves · 1–4 · 100,773

Fig. 15. **Shared-reference neighborhoods.** Reference metadata covers 3,233 of 4,751 papers, from bibliographic data of 9 September 2026, before the corpus was limited to AI papers. Numbers label the seven largest groups of sources cited together.

C DETAILED MECHANISM AND EVIDENCE PROFILES

Coding and populations. This appendix uses the coding of the mechanism cards completed by 28 August 2026, restricted to the systematic corpus (Section 4): 2,847 open-coded papers and 2,891 closed-field papers. Appendix A.6 lists the population behind each profile.

The complete facet catalogue supports the main mechanism analysis (Section 6); complementary evidence profiles support the workflow analysis (Section 5).

C.1 Category Maps and Mechanism Examples

The eighteen category maps cover all 110 mechanism facets in five families. Branches group design operations that systems may combine. Schematics explain the choices; source figures show cited implementations. Variations are recorded implementations, not settings available in every system. Map order and size carry no frequency or effectiveness scale.

Search the full definitions. Links beneath each map open its source mechanisms.

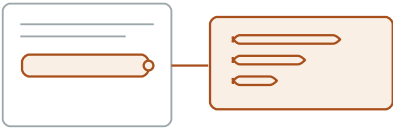
C.1.1 Explanation and visibility. What can a person inspect and understand?

FC-C01 / Explanation and visibility

Explaining system behaviour

Which aspect of the system is made inspectable?

Makes a system's outputs, behaviour, uncertainty or current state inspectable, so people can understand what it is doing and judge its results.



DESIGN CHOICES

RETAINED DISTINCTIONS

A particular result

FC001

output rationale exposed

FC002

feature attribution for an output

Process and sequence

FC006

live process-state visibility

FC008

stepwise reasoning or execution trace

Model and internals

FC003

global model-behaviour summary

FC007

internal or latent state surfaced

Reliability and limits

FC004

uncertainty and confidence exposed

FC005

capability and knowledge-boundary disclosure

Recorded form · FC001: Rationale for a specific machine output (VA001).

Recorded form · FC002: Feature or factor contribution to an outcome (VA002).

Fig. 16. Explaining system behaviour. Which aspect of the system is made inspectable? Separate the explanation of one result, the running process, the model across cases, and the limits of what it can establish. Category definitions; Browse category mechanisms. Recorded forms: Rationale for a specific machine output; Feature or factor contribution to an outcome.

FC-C01 / SOURCE ILLUSTRATION**Explaining system behaviour****PAPER ILLUSTRATION · FC006**

1. Intended route overlaid on the road 2. Detected obstacle marked in the scene

Live overlays expose the vehicle's planned route and detected objects in the road scene.

ExplAIn Yourself! Transparency for Positive UX in Autonomous Driving · Figure 2, PDF page 4

Schneider et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 17. Explaining system behaviour: source illustration. Figure 2, PDF page 4 in [127]. Live overlays expose the vehicle's planned route and detected objects in the road scene. Inspect the illustrated mechanism.

FC-C02 / Explanation and visibility

Sources and evidence

How can a result be traced or checked?

Connects a result to its sources, supporting evidence or underlying cases, so people can trace where it came from and check its basis.



DESIGN CHOICES

RETAINED DISTINCTIONS

Origin and derivation

FC009
output-to-source provenance link

FC012
analytic-history provenance capture

Underlying cases

FC010
aggregate-to-instance drill-down

Evidence for a claim

FC011
evidence presented for verification

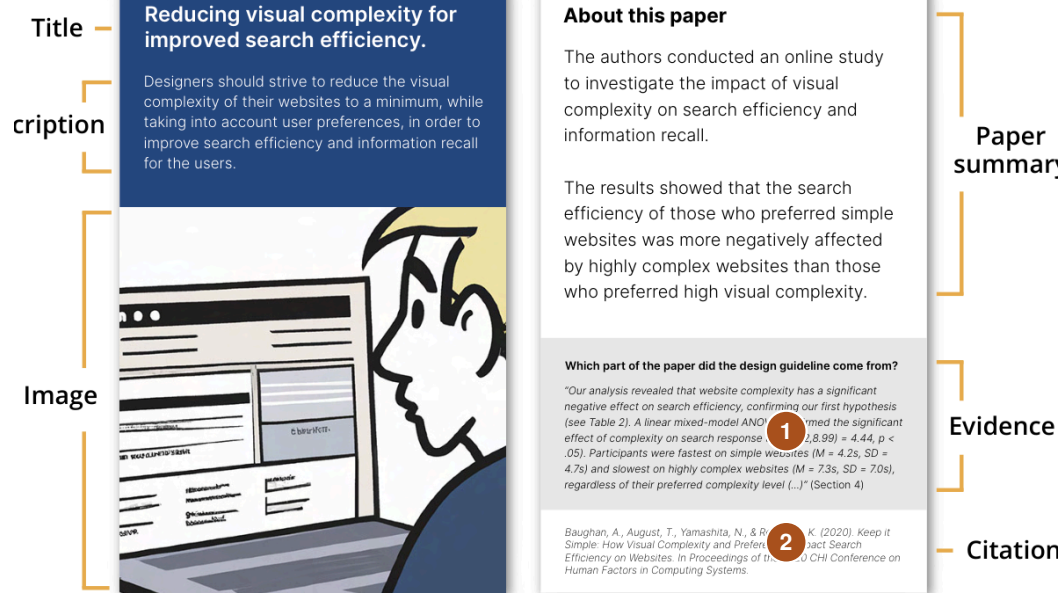
Recorded form · FC010: Drill-down from summary to underlying instances (VA012).

Fig. 18. Sources and evidence. How can a result be traced or checked? Tracing origin and derivation, opening underlying cases, and assessing evidence answer different checking questions. Category definitions; Browse category mechanisms. Recorded forms: Drill-down from summary to underlying instances.

FC-C02 / SOURCE ILLUSTRATION

Sources and evidence

PAPER ILLUSTRATION · FC009



1. Supporting passage 2. Citation back to the paper

The design card places a supporting source passage and its citation beside the implication.

From Paper to Card: Transforming Design Implications with Generative AI · Figure 1, PDF page 2

Shin et al.; CC BY; cropped; numbered annotations added by this review.

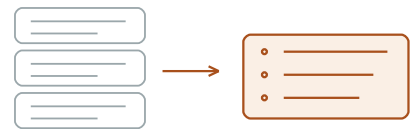
Fig. 19. Sources and evidence: source illustration. Figure 1, PDF page 2 in [133]. The design card places a supporting source passage and its citation beside the implication. Inspect the illustrated mechanism.

FC-C03 / Explanation and visibility

Summarising and labelling content

How is content condensed, interpreted or recast?

Condenses or interprets content through summaries, labels, profiles or a change of representation, so people can grasp what it means.



DESIGN CHOICES

RETAINED DISTINCTIONS

Condense content

FC107
summarisation and condensation

Characterise an item

FC105
machine-generated semantic
labelling

FC109
entity or concept profile

Change its form

FC108
semantic transformation into another form

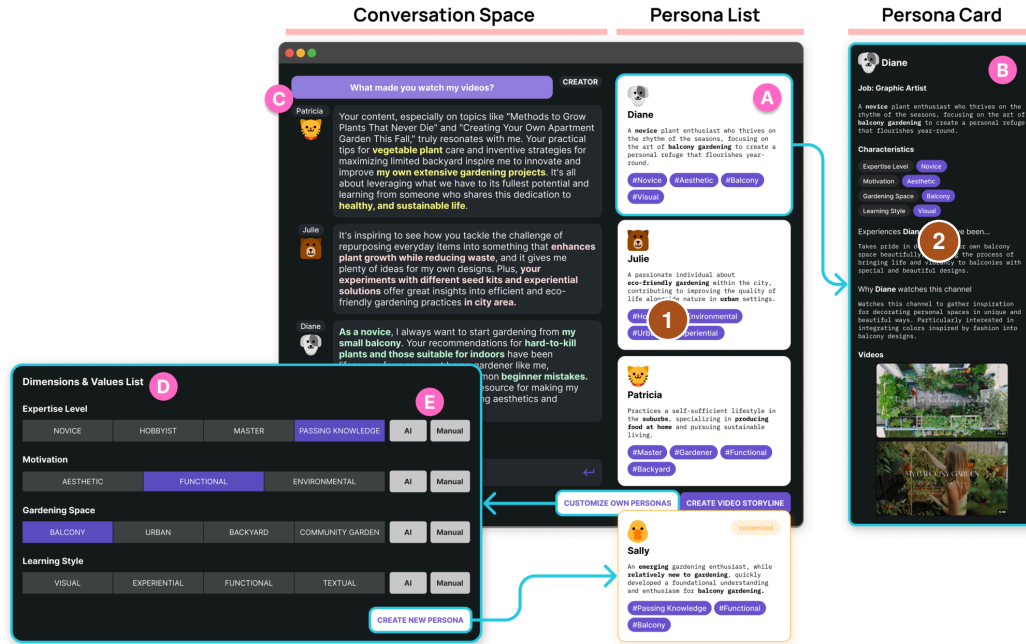
Recorded form · FC107: Condensed summary of a single item (VA097).

Fig. 20. Summarising and labelling content. How is content condensed, interpreted or recast? Distinguish reducing a body of content, characterising an item, and changing representational kind. Category definitions; Browse category mechanisms. Recorded forms: Condensed summary of a single item.

FC-C03 / SOURCE ILLUSTRATION

Summarising and labelling content

PAPER ILLUSTRATION · FC109



1. Compact audience profiles 2. Expanded persona details

Audience segments become profile cards, with detailed experiences and motivations available on selection.

Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas · Figure 2, PDF page 6

Choi et al.; CC BY; cropped; numbered annotations added by this review.

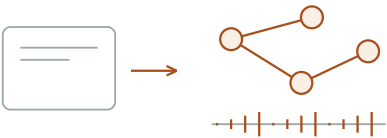
Fig. 21. Summarising and labelling content: source illustration. Figure 2, PDF page 6 in [31]. Audience segments become profile cards, with detailed experiences and motivations available on selection. Inspect the illustrated mechanism.

FC-C04 / Explanation and visibility

Visual and sensory representation

Through which form can a person perceive meaning?

Expresses meaning through visual form, language, sound, touch or physical action, including translations and adaptations that make it accessible.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Perceptual carrier	FC085 visual encoding and graphical mapping	FC083 non-visual sensory encoding	FC086 embodied or physical enactment
Translation of meaning	FC084 cross-modal translation	FC089 verbal description of perceptual content	
Adaptation and reframing	FC088 accessible or simplified presentation	FC087 metaphorical or symbolic re-representation	

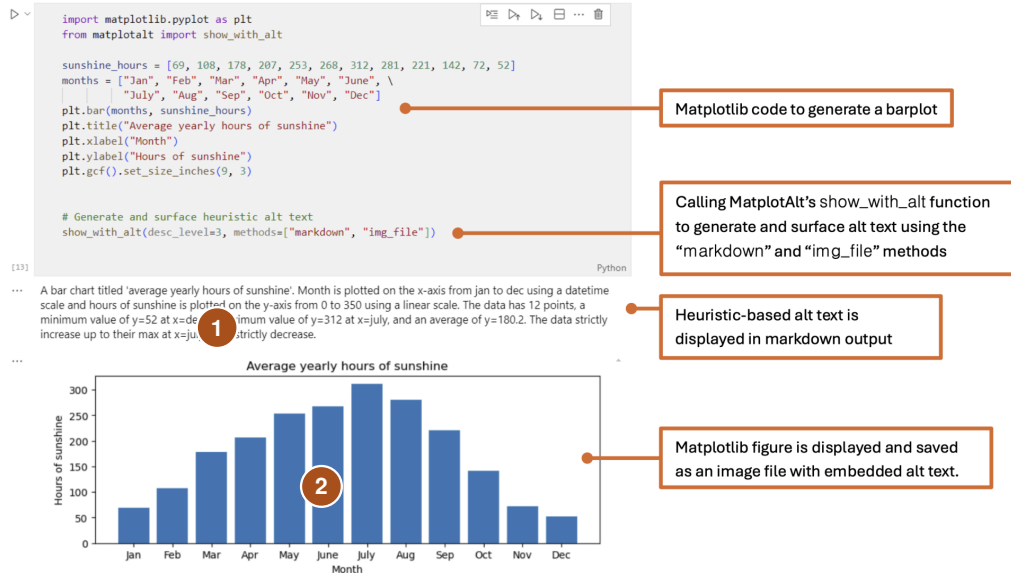
Recorded form · FC083: Spoken or audio rendering of content (VA071).

Fig. 22. Visual and sensory representation. Through which form can a person perceive meaning? Separate the perceptual carrier, translation between carriers, and adaptations that change how the meaning can be grasped. Category definitions; Browse category mechanisms. Recorded forms: Spoken or audio rendering of content.

FC-C04 / SOURCE ILLUSTRATION

Visual and sensory representation

PAPER ILLUSTRATION · FC088



1. Description surfaced as text 2. The same chart remains visible

A notebook presents a chart and its generated description together; the library also supports alternative output placements.

Matplotlib: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks · Figure 1, PDF page 1

Nylund et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 23. Visual and sensory representation: source illustration. Figure 1, PDF page 1 in [109]. A notebook presents a chart and its generated description together; the library also supports alternative output placements. Inspect the illustrated mechanism.

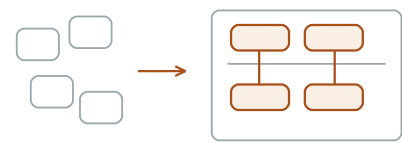
C.1.2 Organisation and comparison. How can a person organise and compare material?

FC-C05 / Organisation and comparison

Organising and structuring

Which relationships give material its structure?

Arranges information into relationships, groups, levels or sequences, so people can understand its structure and move between the whole and its parts.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Links and containment	FC015 relational or network map	FC013 hierarchical or nested organisation	
Groups and categories	FC016 similarity grouping and clustering	FC017 categorisation scheme or controlled vocabulary	FC106 human semantic coding and categorisation
Whole, parts and detail	FC014 overview or aggregate gist	FC018 decomposition into parts	FC019 multiscale detail and semantic zoom
Position and alignment	FC020 temporal arrangement of material	FC021 spatial arrangement of material	FC022 cross-representation correspondence
Quantitative structure	FC023 distributional or statistical structure display		

Recorded form · FC015: Relational graph or dependency network (VA019).

Fig. 24. Organising and structuring. Which relationships give material its structure? Relations, category membership, granularity, position and statistical structure organise different aspects of the same material and may be combined. Category definitions; Browse category mechanisms. Recorded forms: Relational graph or dependency network.

FC-C05 / SOURCE ILLUSTRATION

Organising and structuring

PAPER ILLUSTRATION · FC018



6. Research question node 7. Linked solution node

An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes.

Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback · Figure 4, PDF page 8

Pu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 25. Organising and structuring: source illustration. Figure 4, PDF page 8 in [117]. An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes. Inspect the illustrated mechanism.

FC-C06 / Organisation and comparison

Comparing alternatives

What is placed in a common comparison frame?

Places alternatives, cases or states in a common frame so people can examine their similarities, differences and trade-offs.

	A	B

DESIGN CHOICES

RETAINED DISTINCTIONS

Separate candidates

FC024
side-by-side alternative comparison

Changed states or versions

FC025
contrastive or counterfactual comparison

FC028
change-over-time comparison

Standards and populations

FC026
benchmark or peer-referenced comparison

FC027
subgroup or population comparison

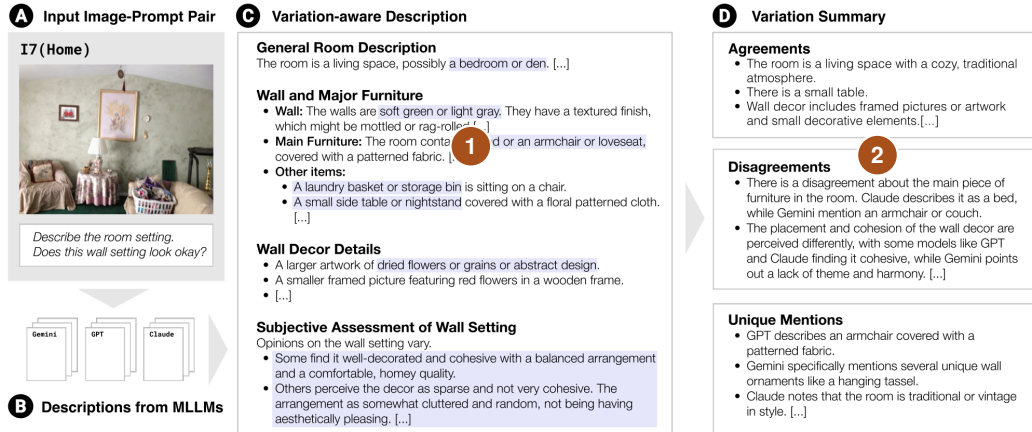
Recorded form · FC024: Side-by-side comparison of alternatives (VA023).
Recorded form · FC025: Before-and-after or version comparison (VA025).

Fig. 26. Comparing alternatives. What is placed in a common comparison frame? The comparand distinguishes candidate selection, variation of the same object, and comparison against groups or reference points. Category definitions; Browse category mechanisms. Recorded forms: Side-by-side comparison of alternatives; Before-and-after or version comparison.

FC-C06 / SOURCE ILLUSTRATION

Comparing alternatives

PAPER ILLUSTRATION · FC024



1. Detailed merged description 2. Agreements and disagreements

The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary.

Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions · Figure 2, PDF page 5

Chen et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 27. Comparing alternatives: source illustration. Figure 2, PDF page 5 in [26]. The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary. Inspect the illustrated mechanism.

FC-C07 / Organisation and comparison

Search, filtering and ranking

How is relevant material found and brought into view?

Helps people find relevant material and control what is in view through search, navigation, filters, rankings and scope choices.

The diagram illustrates a search interface. It features a search bar at the top with a magnifying glass icon. Below the search bar are three stacked filter buttons. The top filter button is highlighted with an orange border, while the two buttons below it have standard grey borders.

DESIGN CHOICES

RETAINED DISTINCTIONS

Define the request

FC030
query construction and search specification

FC033
analytic scope setting

Select and order

FC029
relevance filtering of a corpus

FC032
ranked or prioritised surfacing

Navigate and look up

FC034
navigation and wayfinding through structure

FC031
on-demand lookup of related information

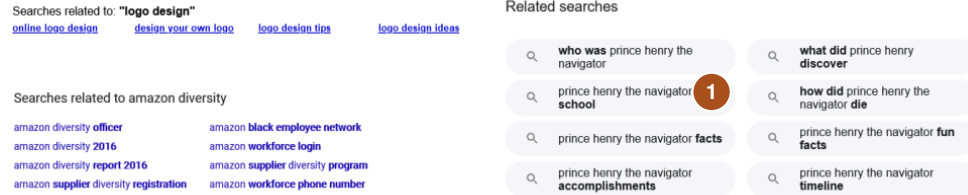
Recorded form · FC030: Filter, query or search over a set (VA028).

Fig. 28. Search, filtering and ranking. How is relevant material found and brought into view? Defining the request, selecting or ordering results, and moving through material are distinct retrieval operations. Category definitions; Browse category mechanisms. Recorded forms: Filter, query or search over a set.

FC-C07 / SOURCE ILLUSTRATION

Search, filtering and ranking

PAPER ILLUSTRATION · FC030



1. Follow-on query formulations

Related-search suggestions provide alternative formulations for continuing a search.

From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features · Figure 16, PDF page 5

Oliveira et al.; CC BY; cropped; numbered annotations added by this review.

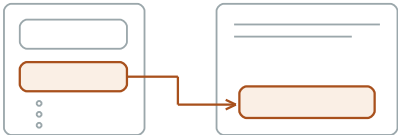
Fig. 29. Search, filtering and ranking; source illustration. Figure 16, PDF page 5 in [110]. Related-search suggestions provide alternative formulations for continuing a search. Inspect the illustrated mechanism.

FC-C08 / Organisation and comparison

Notes, history and memory

What is kept available for continuing the work?

Keeps information and work history available through notes, annotations, memory aids and replay, so people can revisit and continue their work.



DESIGN CHOICES

RETAINED DISTINCTIONS

Express and capture

FC077
externalising internal structure into an artifact

FC080
capture of activity as it happens

Mark and arrange

FC078
annotation onto material

FC082
reorganisable working workspace

Recall and revisit

FC079
external memory and recall cue

FC081
replay and history navigation

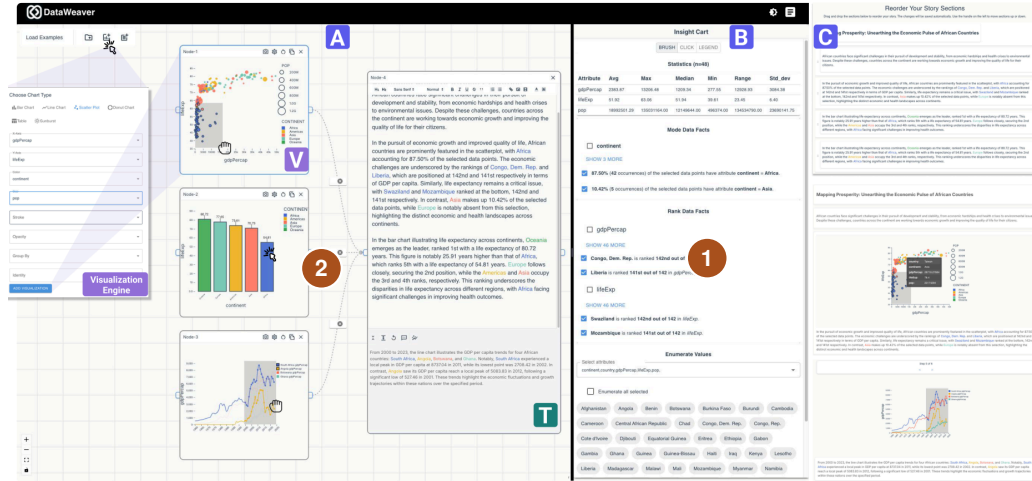
Recorded form · FC078: Annotation and comment capture on content (VA048).

Fig. 30. Notes, history and memory. What is kept available for continuing the work? Separate making or capturing a record, arranging material at hand, and returning to an earlier state. Category definitions; Browse category mechanisms. Recorded forms: Annotation and comment capture on content.

FC-C08 / SOURCE ILLUSTRATION

Notes, history and memory

PAPER ILLUSTRATION · FC082



1. Facts held in the Insight Cart 2. Connected story components

An Insight Cart holds candidate facts for later selection into a connected data story.

DATAWEAVER: Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text · Figure 2, PDF page 5

Fu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 31. Notes, history and memory: source illustration. Figure 2, PDF page 5 in [43]. An Insight Cart holds candidate facts for later selection into a connected data story. Inspect the illustrated mechanism.

C.1.3 User control and expression. What can a person express, change or try?

FC-C09 / User control and expression

Goals and constraints

How does a person specify what the system should do?

Lets people state what they want through goals, rules, criteria, examples or structured inputs that the system can act on.

DESIGN CHOICES

RETAINED DISTINCTIONS

Words and demonstrations

FC053
natural-language or conversational specification

FC054
example or demonstration specification

Structured and executable input

FC057
structured input form or schema

FC058
authoring a formal or executable representation

Conditions and criteria

FC055
explicit constraint or rule specification

FC056
criteria and rubric specification

Recorded form · FC055: Rule, criterion or rubric authoring (VA042).

Fig. 32. Goals and constraints. How does a person specify what the system should do? Distinguish conveying intent, providing a structured or executable specification, and declaring constraints or evaluation criteria. Category definitions; Browse category mechanisms. Recorded forms: Rule, criterion or rubric authoring.

FC-C09 / SOURCE ILLUSTRATION

Goals and constraints

PAPER ILLUSTRATION · FC054

☐ Show only Comments labeled as "Remove" 30 / 30 labeled examples; 14 example(s) labeled as "Remove".

Those people who pull out guns just because they get upset are called gang banger thug losers. 99% of their guns are illegally possessed. Legal gun owners aren't taking out their guns during an argument at parties.

So I should get rid of my gun because some idiot killed.

Load More Examples Create your Filter

1. Keep or remove this example 2. Train the filter from labels

Situated keep/remove judgments supply examples of the content a person wants the classifier to retain.

End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting · Figure 2, PDF page 6

Wang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 33. Goals and constraints: source illustration. Figure 2, PDF page 6 in [145]. Situated keep/remove judgments supply examples of the content a person wants the classifier to retain. Inspect the illustrated mechanism.

FC-C10 / User control and expression

Editing and steering

Where can a person intervene in the work?

Lets people change a representation, revise an output or redirect a process through edits, settings, parameters and reversible actions.



DESIGN CHOICES

RETAINED DISTINCTIONS

Object and output

FC047
direct manipulation of a semantic object

FC048
editable machine output

Settings and quantities

FC050
configuration of system behaviour

FC049
continuous parameter or dial control

Direction and commitment

FC051
steering an ongoing process

FC052
reversibility and safe experimentation

Recorded form · FC049: Parameter, threshold or weight controls (VA039).

Fig. 34. Editing and steering. Where can a person intervene in the work? Editing an object, configuring behaviour, steering the current process and controlling commitment operate at different points. Category definitions; Browse category mechanisms. Recorded forms: Parameter, threshold or weight controls.

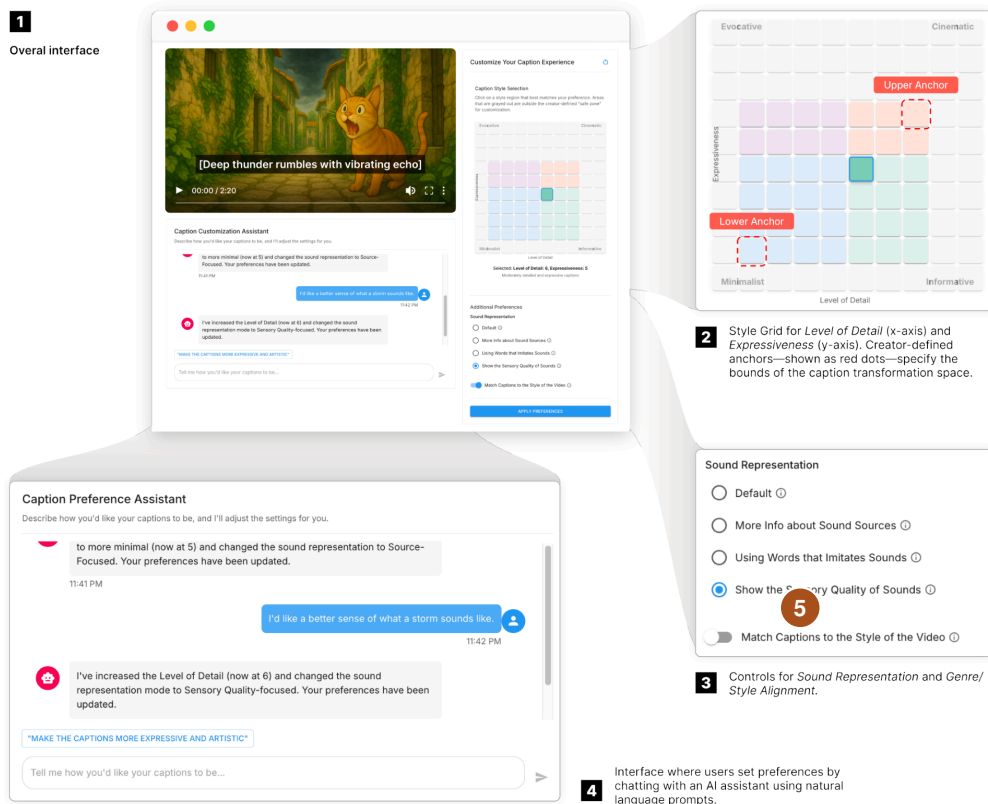
FC-C10 / SOURCE ILLUSTRATION

Editing and steering

PAPER ILLUSTRATION · FC050

1

Overall interface



5. Choose how the sound is represented

Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material.

CapTune: Adapting Non-Speech Captions With Anchored Generative Models · Figure 4, PDF page 8

Huang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 35. Editing and steering: source illustration. Figure 4, PDF page 8 in [60]. Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material. Inspect the illustrated mechanism.

FC-C11 / User control and expression
Generating and choosing alternatives

How are possibilities offered and shaped?

Provides candidate outputs, continuations or a space of possibilities that people can explore, constrain and select from.



DESIGN CHOICES

RETAINED DISTINCTIONS

Present a choice space

FC035
candidate or option set to choose from

FC038
option-space landscape or design space

Generate or continue

FC036
generated alternatives to explore

FC110
completion and continuation of content

Bound generation

FC037
user-constrained generation

Recorded form · FC036: Gallery of generated alternatives to choose among (VA033).

Recorded form · FC110: Completion or continuation of what the person began (VA036).

Fig. 36. Generating and choosing alternatives. How are possibilities offered and shaped? Separate presenting a choice space, producing new content, and bounding the generation process. A continuation is not necessarily a set of alternatives. Category definitions; Browse category mechanisms. Recorded forms: Gallery of generated alternatives to choose among; Completion or continuation of what the person began.

FC-C11 / SOURCE ILLUSTRATION

Generating and choosing alternatives

PAPER ILLUSTRATION · FC110

Double-click the snippet to edit it directly. **THIS IS SUGGESTED TEXT**

Adjust role

Explain interaction × ⓘ

Adjust formatting

heading drophead body **note**

Get recommendations

Highlight interactive elements and guide the audience's interaction flow using moderately complex sentences. Use imperative sentences to instruct users on actions they can perform. Depending on the complexity, aim for 10-25 words, a readability level appropriate for someone with a 6th to 10th grade education, and a word variety of 50-70%.

Word count **5** ⓘ

Word variety **40%** ⓘ

Readability score **69** ⓘ

Your text is likely to be understood by someone who has at least a 8th-grade education.

Generate text

↻ Re-generate ✂ Shorten **1** 📄 Simplify

Add context to incorporate into generation

What should we consider when generating this text?

🗑 Delete snippet

1. Regenerate, shorten or simplify

A text snippet exposes explicit actions for generating a revised version while the author continues composing.

Plume: Scaffolding Text Composition in Dashboards · Figure 5, PDF page 11

Lisnic et al.; CC BY; cropped; numbered annotations added by this review.

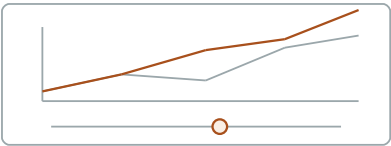
Fig. 37. Generating and choosing alternatives: source illustration. Figure 5, PDF page 11 in [84]. A text snippet exposes explicit actions for generating a revised version while the author continues composing. Inspect the illustrated mechanism.

FC-C12 / User control and expression

What-if exploration and rehearsal

What can be tried before acting in the real setting?

Lets people try hypothetical changes, probe a system's behaviour or practise in a simulated setting and inspect what happens.



DESIGN CHOICES

RETAINED DISTINCTIONS

Probe the system

FC098
system-behaviour probing

Construct a scenario

FC097
scenario construction and what-if exploration

Rehearse an activity

FC096
simulated environment for rehearsal

Recorded form - FC096: Practice, rehearsal or simulation environment (VA058).

Fig. 38. What-if exploration and rehearsal. What can be tried before acting in the real setting? Separate learning how a system responds, constructing a hypothetical world, and rehearsing a person's activity. Category definitions; Browse category mechanisms. Recorded forms: Practice, rehearsal or simulation environment.

FC-C12 / SOURCE ILLUSTRATION

What-if exploration and rehearsal

PAPER ILLUSTRATION · FC096



1. Dialogue history 2. Teacher question entry

Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history.

Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents · Figure 1, PDF page 3

Barrett et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 39. What-if exploration and rehearsal: source illustration. Figure 1, PDF page 3 in [9]. Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history. Inspect the illustrated mechanism.

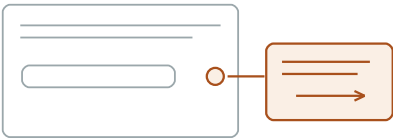
C.1.4 Guidance and feedback. How is the next action or reflection supported?

FC-C13 / Guidance and feedback

Recommendations and guidance

What helps a person decide where to look or what to do?

Directs attention or suggests an action, resource or procedure, including corrective advice, consequence previews and warnings.



DESIGN CHOICES

RETAINED DISTINCTIONS

Notice and hold back

FC043
attention direction and salience
cueing

FC044
risk alert or warning

Preview a
consequence

FC046
prospective consequence preview

Choose an action or
resource

FC040
next-step recommendation

FC042
item or content recommendation

Perform or repair

FC045
procedural how-to instruction

FC041
corrective or remediation guidance

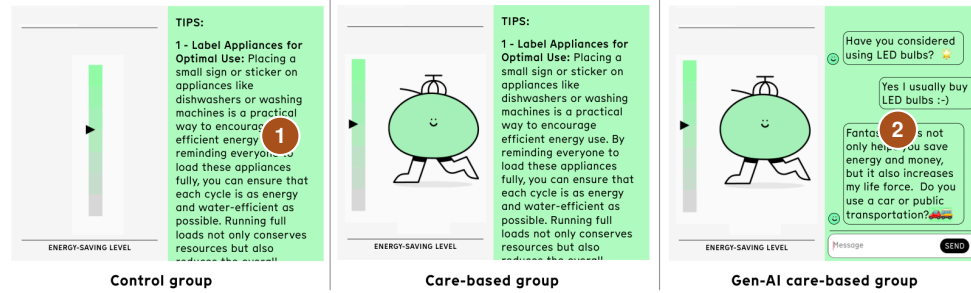
Recorded form · FC046: Preview of the consequence before committing (VA037).

Fig. 40. Recommendations and guidance. What helps a person decide where to look or what to do? Attention and warnings, previews, recommendations, and procedural or corrective guidance contribute different information to an action. Category definitions; Browse category mechanisms. Recorded forms: Preview of the consequence before committing.

FC-C13 / SOURCE ILLUSTRATION

Recommendations and guidance

PAPER ILLUSTRATION · FC040



1. Fixed tips 2. Conversational condition

The study contrasts fixed energy-saving tips with a conversational condition that elicits household context.

Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment · Figure 6, PDF page 6

Berney et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 41. Recommendations and guidance: source illustration. Figure 6, PDF page 6 in [12]. The study contrasts fixed energy-saving tips with a conversational condition that elicits household context. Inspect the illustrated mechanism.

FC-C14 / Guidance and feedback
Feedback and checking

What basis is provided for judging the work?

Shows how work is progressing or how a result meets a goal, criterion or reference. It includes feedback on attempts, problem diagnosis, quality assessments and checks.



DESIGN CHOICES

RETAINED DISTINCTIONS

Attempt and progress	FC065 formative outcome feedback	FC066 progress and status tracking	
Problem, quality and status	FC063 Identifying problems in the work	FC064 evaluative judgment or quality rating	FC067 correctness or status marking
Check against a reference	FC068 verification and checking against a reference		

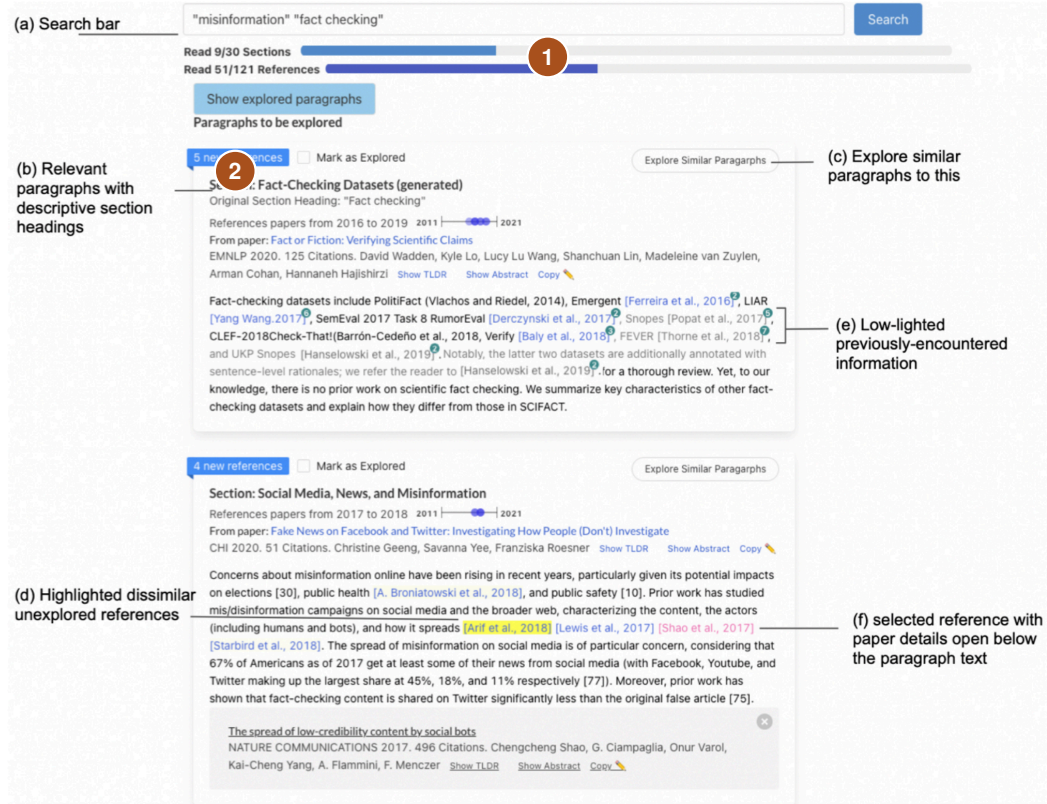
Recorded form · FC063: Error and failure diagnosis (VA007).

Fig. 42. Feedback and checking. What basis is provided for judging the work? Separate feedback about an attempt or progress, appraisals and status marks, and an active check against a reference. Category definitions; Browse category mechanisms. Recorded forms: Error and failure diagnosis.

FC-C14 / SOURCE ILLUSTRATION

Feedback and checking

PAPER ILLUSTRATION · FC066



1. Paragraph and reference coverage 2. Unread-reference count

Persistent coverage bars and local unread-reference counts make progress through the literature visible.

Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections · Figure 1, PDF page 2

Palani et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 43. Feedback and checking: source illustration. Figure 1, PDF page 2 in [112]. Persistent coverage bars and local unread-reference counts make progress through the literature visible. Inspect the illustrated mechanism.

FC-C15 / Guidance and feedback

Instruction and paced help

What help is available, and when is it available?

Provides explanations, examples and orientation, and controls when or how much help is available through staged disclosure, learning paths or changing support.



DESIGN CHOICES

RETAINED DISTINCTIONS

Explain, exemplify and orient

FC101
conceptual explanation of domain meaning

FC102
worked example or exemplar

FC103
orientation and onboarding

Change amount or release

FC099
graduated or faded support

FC100
progressive disclosure of detail

FC039
deliberate restriction or withdrawal of support

Sequence learning sessions

FC104
curriculum and learning-path structuring

Recorded form · FC099: Graduated help that escalates or fades by response (VA093).

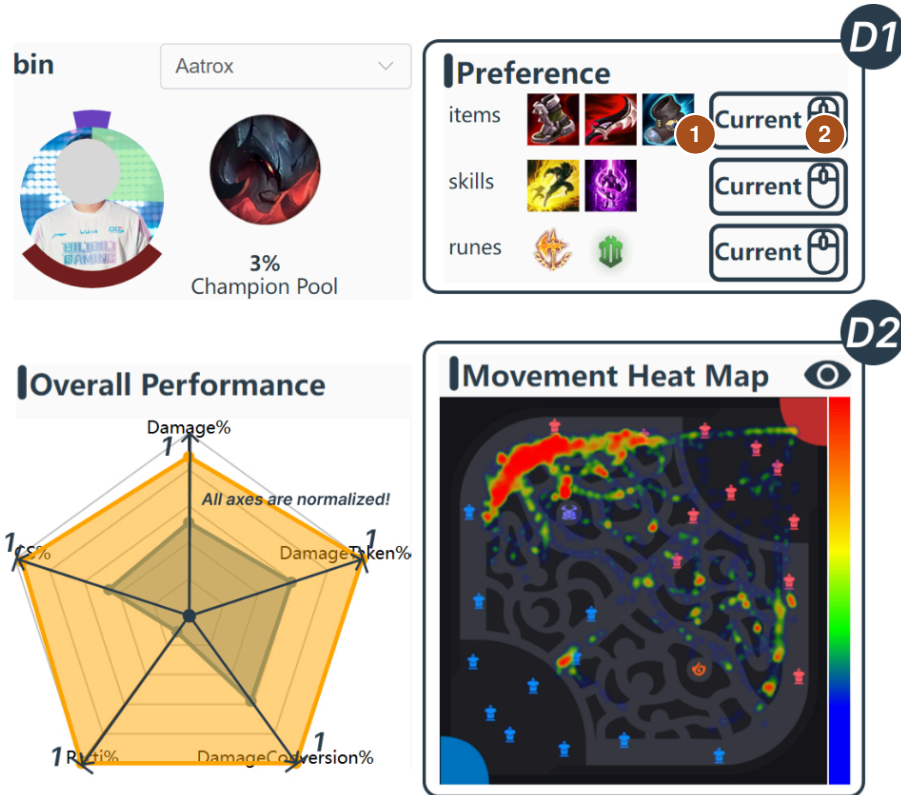
Recorded form · FC102: Worked or contrastive examples as instruction (VA057).

Fig. 44. Instruction and paced help. What help is available, and when is it available? Instructional content, the amount or timing of support, and progression across sessions are distinct design choices. Fading is not automatically evidence of learning. Category definitions; Browse category mechanisms. Recorded forms: Graduated help that escalates or fades by response; Worked or contrastive examples as instruction.

FC-C15 / SOURCE ILLUSTRATION

Instruction and paced help

PAPER ILLUSTRATION · FC102



1. Historical equipment choices 2. Switch to current choices

Historical equipment, skill and rune choices can be inspected beside controls revealing the current match.

StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports · Figure 8, PDF page 12

Ma et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 45. Instruction and paced help: source illustration. Figure 8, PDF page 12 in [89]. Historical equipment, skill and rune choices can be inspected beside controls revealing the current match. Inspect the illustrated mechanism.

FC-C16 / Guidance and feedback
Structuring the workflow

How are stages, responsibilities and continuity arranged?

Organises work into stages, loops, roles and decision points, with ways to coordinate actions, hand work over, recover from problems and resume later.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Sequence and recurrence	—	FC069 staged or phased workflow	FC070 iterative revision loop
Responsibility and routing	—	FC071 human decision checkpoint or gate	FC072 division of labour across agents
			FC074 orchestration and routing of actions
Continuity and recovery	—	FC073 handoff of work state	FC075 interaction repair and recovery
			FC076 resumable and persistent work state

Recorded form · FC075: System move that clarifies, confirms or repairs (VA038).

Fig. 46. Structuring the workflow. How are stages, responsibilities and continuity arranged? Separate progression through work, allocation of actions and authority, and continuity across interruptions or handoffs. Category definitions; Browse category mechanisms. Recorded forms: System move that clarifies, confirms or repairs.

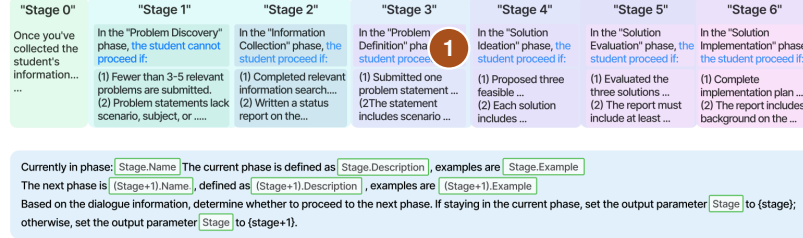
FC-C16 / SOURCE ILLUSTRATION

Structuring the workflow

PAPER ILLUSTRATION · FC074

Controller Agent Prompts

[Stage Decision Prompt]



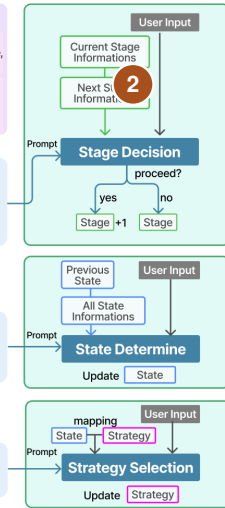
[State Determine Prompt]

If not moving to the next phase, the student's status may be: `State.Name` `State.Description`
 The student's previous status list is `State`. Remove all '0' and '1', and based on the new dialogue record, update and output the student's status list state.
 If moving to the next phase, the student's status will be "start," and update and output the student's status list as `state=[0]`.

[Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: `State.Name` `State.StrategyMapping` `Strategy.name` `Strategy.Description`
 Based on the dialogue information, output the most urgent problem ID as `urge_state` and the corresponding best strategy ID as `strategy`

Controller Agent Pipeline



1. Stage-specific completion criteria 2. Decision to advance or remain

Stage criteria and a controller decision determine whether the creative problem-solving workflow advances.

Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process · Figure 2, PDF page 9

Zha et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 47. Structuring the workflow: source illustration. Figure 2, PDF page 9 in [158]. Stage criteria and a controller decision determine whether the creative problem-solving workflow advances. Inspect the illustrated mechanism.

FC-C17 / Guidance and feedback

Questions and reflection prompts

Which part of a person's thinking is elicited?

Asks people to express their goals, questions, explanations or reflections, making their own reasoning available to the work.

?

DESIGN CHOICES

RETAINED DISTINCTIONS

Needs and intentions

FC059

intent and need elicitation

Account and reasoning

FC062

narrative and reasoning elicitation

Questions and hypotheses

FC061

inquiry and hypothesis prompting

Reflection on thinking

FC060

reflective and metacognitive prompting

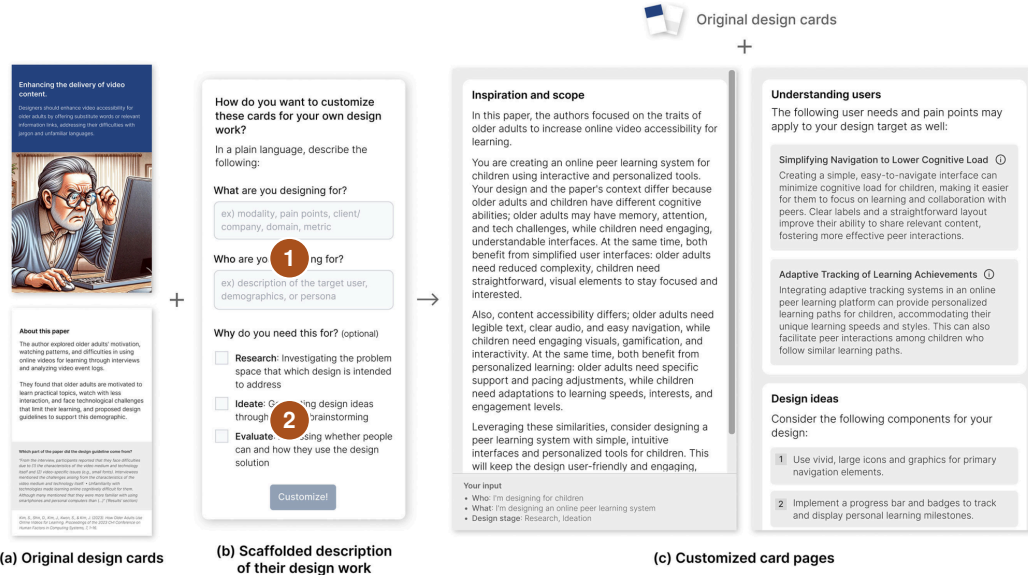
Recorded form · FC060: Reflection or self-assessment prompt (VA056).

Fig. 48. Questions and reflection prompts. Which part of a person's thinking is elicited? The object of the prompt distinguishes needs, an account of reasoning, inquiry about the material, and reflection on one's own thinking. Category definitions; Browse category mechanisms. Recorded forms: Reflection or self-assessment prompt.

FC-C17 / SOURCE ILLUSTRATION

Questions and reflection prompts

PAPER ILLUSTRATION · FC059



1. Who and what is being designed 2. Select the design stage

A guided form elicits design context before the system customizes research cards.

What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts · Figure 1, PDF page 7

Shin et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 49. Questions and reflection prompts: source illustration. Figure 1, PDF page 7 in [132]. A guided form elicits design context before the system customizes research cards. Inspect the illustrated mechanism.

C.1.5 *Collaboration and perspective.* How can people share work and perspectives?**FC-C18 / Collaboration and perspective****Shared views and perspectives****How do contributions and viewpoints become shared?**

Makes people's contributions and viewpoints available to one another through shared representations, structured discussion, role framing and control over what is disclosed.

**DESIGN CHOICES****RETAINED DISTINCTIONS****Shared work and awareness**

FC090
shared co-constructed representation

FC091
visibility of others' state and judgment

Viewpoints and exchange

FC092
perspective-taking and viewpoint framing

FC094
role or persona framing of the interaction

FC093
structured deliberation and perspective exchange

Control of disclosure

FC095
disclosure and privacy control

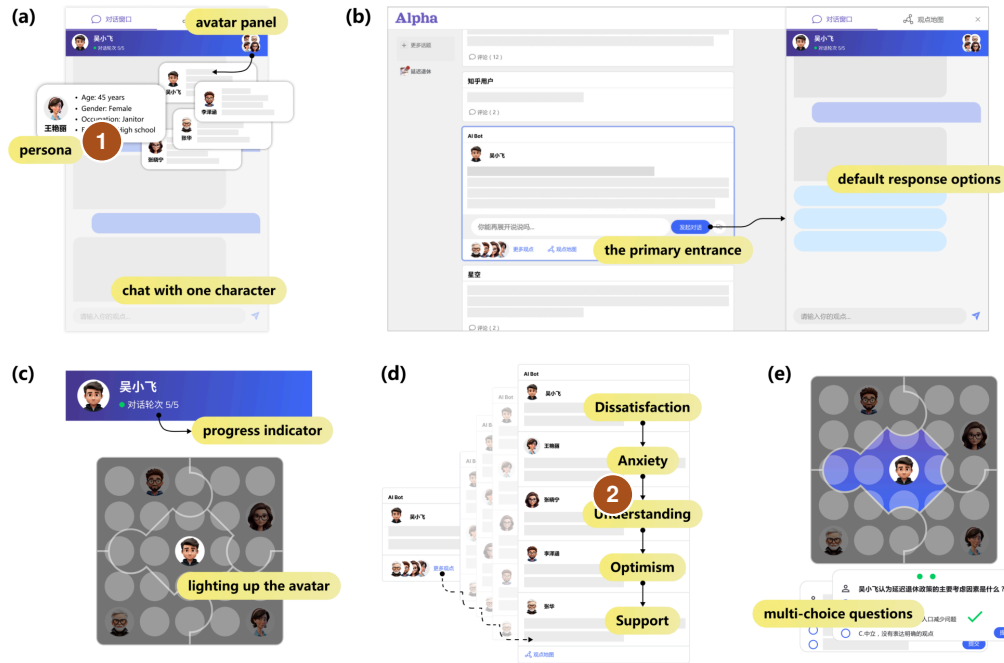
Recorded form · FC090: Shared workspace for joint work (VA065).

Fig. 50. Shared views and perspectives. How do contributions and viewpoints become shared? Separate shared artifacts and awareness, the framing and exchange of perspectives, and control of disclosure. Category definitions; Browse category mechanisms. Recorded forms: Shared workspace for joint work.

FC-C18 / SOURCE ILLUSTRATION

Shared views and perspectives

PAPER ILLUSTRATION · FC092



1. Distinct stakeholder profiles 2. Progressive viewpoint sequence

Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure.

See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles · Figure 3, PDF page 7

Zhang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 51. Shared views and perspectives: source illustration. Figure 3, PDF page 7 in [162]. Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure. Inspect the illustrated mechanism.

C.2 Complementary Evidence Profiles

The learning-by-family matrix uses the open-code classification and its 2,761-paper matched population. It is distinct from the closed-field learning count of 301 among 2,891 included papers.

Claim support and developmental risk indicators

Each row is a mutually exclusive coding over the same 2,891 included cards.

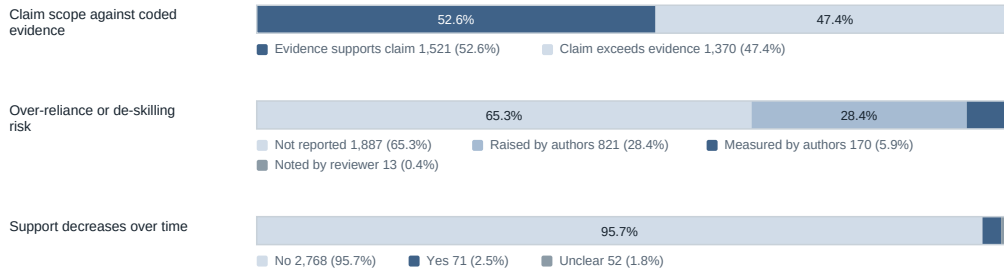


Fig. 52. **Claim scope exceeds coded evidence in nearly half of included cards, over-reliance is raised far more often than it is measured, and decreasing support is rare.** Each row is a mutually exclusive coding over the same 2,891 included cards, and every segment prints its own count and share. A claim-evidence gap records the distance between a card's stated claim and its linked evidence; it is not an assessment of the underlying paper. Examples, one or two cards per value: claim within evidence: InstructPipe [164].

Support profiles within learning-evidence groups

share of the row's papers carrying each family; 2,761 papers coded in both fields

	what the interface makes reachable				
	Organisation and comparison	Guidance and feedback	Explanation and visibility	User control and expression	Collaboration and perspective
learning evidence					
Learning not measured n = 1,334 papers	73.8% 985	66.9% 893	64.2% 856	52.5% 701	11.4% 152
A proxy measured instead n = 1,039 papers	71.3% 741	74.1% 770	61.5% 639	50.0% 520	14.6% 152
Learning measured n = 258 papers	60.9% 157	82.9% 214	57.4% 148	41.5% 107	14.3% 37
Learning self-reported only n = 130 papers	71.5% 93	84.6% 110	63.8% 83	54.6% 71	20.0% 26
all papers in the matrix n = 2,761 papers	71.6%	72.0%	62.5%	50.7%	13.3%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 53. **Guidance and feedback appear in 83% of capability-measured papers.** Rows: the learning-evidence families, including the papers that measured no capability, with the number of papers in each. Columns: the facet families. Each cell is the share of the row's papers carrying the column family, with the count; colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

C.3 Context and Evidence Profiles

The 2,891 closed-field papers record 2,071 empirical, 691 artifact, 123 methodological, four dataset, and two theoretical contributions. Contribution labels do not identify research methods or target users; study designs and participants are reported in Section 5.2.

In the design space, the domain mix shows no simple annual shift: technical, analytic, and creative professions account for 54.8% in 2019 and 55.5% in 2025, ranging from 53.4% to 61.6% across these years. Annual totals also reflect reading coverage and the incomplete 2026 publication window, so they do not estimate field-wide growth.

What is made reachable, against what the person still does

share of the row's papers in each column; 2,311 papers coded in both fields

what the interface makes reachable	what the person still does			
	The task the tool did not do	New work the tool created	The whole task kept by the person	Final authority kept by design
Organisation and comparison n = 1,639 papers	52.2% 856	29.1% 477	11.4% 187	7.1% 117
Guidance and feedback n = 1,679 papers	49.9% 838	30.1% 506	11.1% 187	8.8% 147
Explanation and visibility n = 1,452 papers	53.2% 772	28.0% 407	10.1% 146	8.7% 126
User control and expression n = 1,170 papers	42.0% 491	35.8% 419	14.2% 166	7.9% 93
Collaboration and perspective n = 311 papers	56.3% 175	19.0% 59	12.9% 40	11.9% 37
all papers in the matrix n = 2,311 papers	51.4%	30.2%	10.1%	8.2%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 54. **The profiles share an ordering, with different shares of new work.** The matrix uses the 2,311 systematic papers with complete facet-value and remaining-work assignments. Rows: the facet families, with the number of papers carrying each. Columns: what the person still does. Each cell is the share of the row's papers whose remaining work falls in the column family, with the count. Papers can belong to several mechanism-family rows; each has one remaining-work summary category in this analysis. Colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

Study design is recorded for 2,071 papers: 691 within-subject designs (33.4%), 328 single-condition usability studies (15.8%), and 119 field deployments (5.7%). The 2,072 entries include one paper with two designs; percentages use papers. Usability studies reveal practical difficulties; deployments can expose work missed by short comparisons.

The participant field also covers 2,071 papers: mixed groups account for 540 (26.1%), target practitioners 499 (24.1%), and crowdworkers 251 (12.1%); 382 (18.4%) are unspecified. Participants need not represent a system's intended users, and missing descriptions do not establish broad applicability.

C.3.1 Most process shifts rest on an author's attribution. Of 8,045 process shifts across 2,891 papers, 6,102 (75.8%) are author-attributed, 1,746 (21.7%) use measured contrasts, and 197 are reviewer inferences. Baselines also differ: 1,940 use a control condition, 869 state none, and the remainder refer to prior practice or an author's assertion. Describing a changed workflow does not establish its effect size.

C.4 Mechanism Families and Remaining Work

The detailed profile complements the control/work comparison in Section 5.2.

D ANNUAL TRENDS AND VENUE RATES

These tests and counts support the comparisons in Section 5 and Section 6. Trends use AI-involved papers with usable codes for each field. Technology follows the AI a paper builds or uses, independently of publication year: generative AI includes hybrids and earlier neural generators; traditional AI/ML includes predictive, discriminative, and symbolic systems. The technology projection covers 3,488 classified papers; 645 remain unassigned. Each annual percentage uses papers with assigned technology and usable coding for that dimension in the same year. Count summaries combine

2019–2026; the annual lines separate the years. Alias disagreements are withheld for the affected field. Adjusted contrasts account for publication year and venue as described below.

D.1 Research-context trends

We model binary context membership with year and venue fixed effects, using venue-clustered standard errors with a finite-sample correction and Student’s t reference distribution. The linear probability coefficient estimates average within-venue change in percentage points per year. Holm adjustment controls the family-wise error rate across the 16 directional tests. Intervals are pointwise 95% confidence intervals. Separate categorical-year tests check temporal variation and departures from a straight trend; these are available with the analysis data. Results describe the coded review frame.

Table 12. Venue-adjusted annual changes in research contexts, in percentage points per year. Primary estimates use 2019–2025; the last column includes available 2026 papers. Sensitivity intervals and adjusted tests are provided online.

Context	Annual change [95% CI]	Adjusted p	With 2026
AI and machine learning practice	-0.59 [-1.17, -0.01]	0.651	-0.64
Data analysis and visualization	-0.35 [-0.90, +0.20]	1.000	-0.46
Software engineering and programming	-0.35 [-0.73, +0.03]	0.881	-0.28
Scientific research practice	+0.35 [-0.03, +0.72]	0.881	+0.39
Information seeking and knowledge work	+0.41 [+0.14, +0.68]	0.065	+0.33
Writing, reading and language	+0.22 [-0.10, +0.55]	1.000	+0.30
Design practice	-0.16 [-0.59, +0.27]	1.000	-0.04
Creative and media production	+0.17 [-0.33, +0.67]	1.000	+0.25
Education and learning	+0.11 [-0.32, +0.54]	1.000	+0.09
Health and clinical care	+0.27 [-0.25, +0.80]	1.000	+0.25
Accessibility and disability	+0.29 [+0.07, +0.52]	0.176	+0.09
Everyday life and personal decisions	-0.28 [-0.60, +0.04]	0.881	-0.08
Online platforms and civic information	-0.09 [-0.48, +0.30]	1.000	-0.03
Physical and embodied activity	-0.18 [-0.61, +0.26]	1.000	-0.11
Business, industry and work	+0.29 [-0.03, +0.60]	0.881	+0.14
Public sector, law and societal systems	-0.04 [-0.20, +0.12]	1.000	-0.11

D.2 Technology comparisons

We compare binary code membership using a linear probability model with technology, categorical publication year, and venue fixed effects. Primary estimates use 2019–2025, and sensitivity models restrict the analysis to 2023–2025 or add partial 2026. Venue-clustered standard errors use a finite-sample correction and a Student t reference with one fewer degree of freedom than the number of venues, and model intervals are pointwise 95% confidence intervals. Holm correction covers all 51 context, activity, family, and category contrasts jointly, with each sensitivity window and venue omission corrected separately. We omit six major venues one at a time: CHI, CSCW, EMNLP, IUI, TVCG, and VIS, each of which supplies at least 5% of the coded papers for workflow, families, or categories in 2019–2025. The choice rests on coding denominators, not outcomes. Weighting checks reweight shared year–venue cells with at least one or at least

Table 13. User control after omitting each major venue and on shared year–venue cells. Gaps are GenAI minus traditional AI/ML in percentage points. Venue-omission gaps are adjusted for year and venue, with pointwise 95% confidence intervals and Holm correction over all 51 outcomes in each fit. Shared-cell gaps are point estimates. Equal-venue gaps also have 95% venue-bootstrap intervals. The one-paper threshold keeps 82.8% of the coded papers, the five-paper threshold 40.8%.

Venue omission	Gap [95% CI]	Holm p
Primary	+13.8 [+8.9, +18.7]	< .001
Omit CHI	+12.8 [+6.9, +18.8]	0.005
Omit CSCW	+13.6 [+8.4, +18.8]	< .001
Omit EMNLP	+14.3 [+9.4, +19.3]	< .001
Omit IUI	+13.9 [+8.5, +19.3]	< .001
Omit TVCG	+12.2 [+7.5, +16.9]	< .001
Omit VIS	+14.5 [+9.7, +19.4]	< .001
Shared-cell weight	At least one per group	At least five per group
Pooled papers	+11.4	+17.3
Within-cell overlap	+12.7	+17.1
Equal cells	+8.5	+16.6
Equal venues	+8.4	+15.1
95% interval	[-2.1, +18.9]	[+9.3, +21.1]
Equal years	+10.3	+18.8

five coded papers per technology by pooled paper count, within-cell overlap, equal cells, equal venues, or equal years. Intervals for equal-venue gaps come from 10,000 resamples of venues.

User control. User control and expression occurs in 634/1,072 GenAI papers (59.1%) and 624/1,381 traditional AI/ML papers (45.2%) in 2019–2025. Omitting each major venue gives gaps from +12.2 to +14.5 points, each with Holm-adjusted support (Table 13). On the 2,031 papers in shared year–venue cells, the gap is +12.7 with overlap weights and +8.4 when venues receive equal weight. Every weighting and cell threshold leaves it positive.

Facets and work codes. The facet and work-code gaps in the main text use the same model on each paper’s latest coding, with a normal reference for intervals. Of the 2,988 technology-classified papers from 2019–2025, 2,532 have facet codes (1,143 GenAI and 1,389 traditional AI/ML), 2,119 have codes for delegated work, and 2,526 have codes for the work people keep. Holm correction runs across the 101 facets coded on at least 40 papers and, separately, across the 24 work codes on at least 30 papers. A gap gets a direction in the text only if it has Holm-adjusted support, keeps its sign under all five weightings at both thresholds, and has an equal-venue interval that excludes zero at one threshold or both (Table 13 and Table 14). Fitted separately for the two models that coded most papers, the control gap is +13.9 (95% CI 8.8 to 19.0; 2,160 papers) and +11.1 (−1.1 to 23.3; 203 papers).

D.3 Temporal change in mechanism families

We distinguish average annual change in family prevalence from annual change in the GenAI–traditional AI/ML difference. Both come from a linear probability model with technology, publication year, their interaction and venue fixed effects. Average change holds the technology mixture fixed at 43.75% GenAI (the original primary cohort’s reference share), separating it from shifts in the composition of the corpus. Positive gap change means movement toward a more positive GenAI difference; it need not enlarge the absolute gap. Venue-clustered intervals use finite-sample correction and a Student t reference. Holm correction covers all ten tests within each window; the 95% confidence intervals are

Table 14. The other gaps given a direction in the text. Gaps are GenAI minus traditional AI/ML in percentage points. Primary gaps are adjusted for year and venue, with 95% confidence intervals. Equal-venue gaps use shared year-venue cells with at least one or at least five papers per technology, with 95% venue-bootstrap intervals. The two thresholds keep 82.7–85.4% and 40.3–46.6% of each coded population. Every gap keeps its sign under all five weightings at both thresholds.

Gap	Primary [95% CI]	Equal venues, papers per group	
		At least one	At least five
Tool generates new artifacts	+16.6 [+12.7, +20.5]	+13.8 [+6.6, +22.3]	+17.7 [+9.3, +29.6]
Tool generates responses and revisions	+9.4 [+5.7, +13.2]	+10.2 [+4.7, +17.1]	+7.6 [+0.8, +14.1]
Tool calculates and models	-11.5 [-15.5, -7.5]	-16.2 [-25.8, -7.2]	-16.2 [-25.6, -7.9]
Tool evaluates and classifies	-9.7 [-14.6, -4.7]	-2.6 [-12.7, +7.9]	-13.2 [-21.5, -2.7]
Tool compares and finds patterns	-6.7 [-9.7, -3.7]	-7.0 [-14.3, -1.1]	-1.2 [-9.5, +8.4]
Generated alternatives to explore	+4.8 [+3.0, +6.6]	+4.3 [+0.9, +8.5]	+1.5 [-1.7, +4.5]
Completion and continuation of content	+3.0 [+1.4, +4.6]	+2.1 [-0.6, +5.3]	+2.1 [+0.6, +3.9]
Person steers the loop	+3.4 [+1.2, +5.6]	+6.7 [+2.4, +11.7]	+0.3 [-5.0, +4.9]
Person improves the system	-5.9 [-7.7, -4.0]	-5.3 [-9.3, -1.8]	-6.4 [-10.9, -2.7]
Organization and comparison (family)	-8.2 [-12.5, -3.8]	-12.2 [-20.4, -4.1]	-8.1 [-17.5, +1.7]
Ranked or prioritized surfacing	-6.5 [-8.3, -4.7]	-3.4 [-7.6, +0.3]	-9.7 [-18.0, -3.7]

Table 15. Mechanism-family trends, 2019–2025: 2,453 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	+1.16 [-0.70, +3.01]	1.000	+1.47 [-0.86, +3.80]	1.000
Organisation and comparison	+0.08 [-1.70, +1.86]	1.000	+0.49 [-2.04, +3.01]	1.000
User control and expression	-1.02 [-2.47, +0.42]	0.961	+0.53 [-1.12, +2.17]	1.000
Guidance and feedback	+1.61 [+0.72, +2.50]	0.007	+1.91 [-0.38, +4.20]	0.692
Collaboration and perspective	+1.13 [+0.53, +1.73]	0.005	+2.15 [-0.02, +4.32]	0.414

Table 16. Mechanism-family trends, 2023–2025: 1,391 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	-0.34 [-5.73, +5.06]	1.000	+1.44 [-6.17, +9.05]	1.000
Organisation and comparison	-2.86 [-6.65, +0.94]	0.952	-2.86 [-9.68, +3.96]	1.000
User control and expression	-3.41 [-6.55, -0.27]	0.273	-0.41 [-6.16, +5.34]	1.000
Guidance and feedback	+1.92 [-2.69, +6.54]	1.000	+3.49 [-3.80, +10.77]	1.000
Collaboration and perspective	+2.84 [+1.27, +4.41]	0.007	+6.28 [+1.67, +10.89]	0.080

pointwise. These are linear summaries of coded design prevalence, not tests of learning or causal technology effects. Partial 2026 is excluded.

D.4 Annual distributions

Research contexts: volume and annual share

3,307 coded papers with assigned AI technology · categories overlap

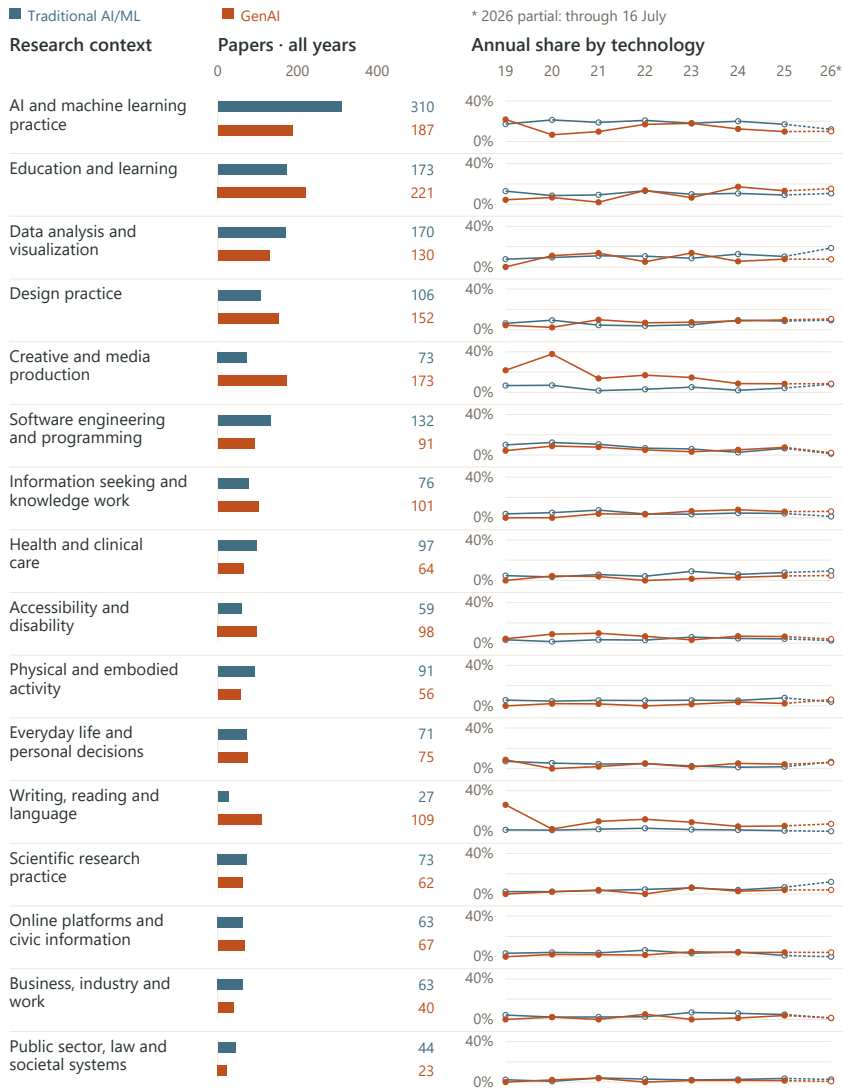
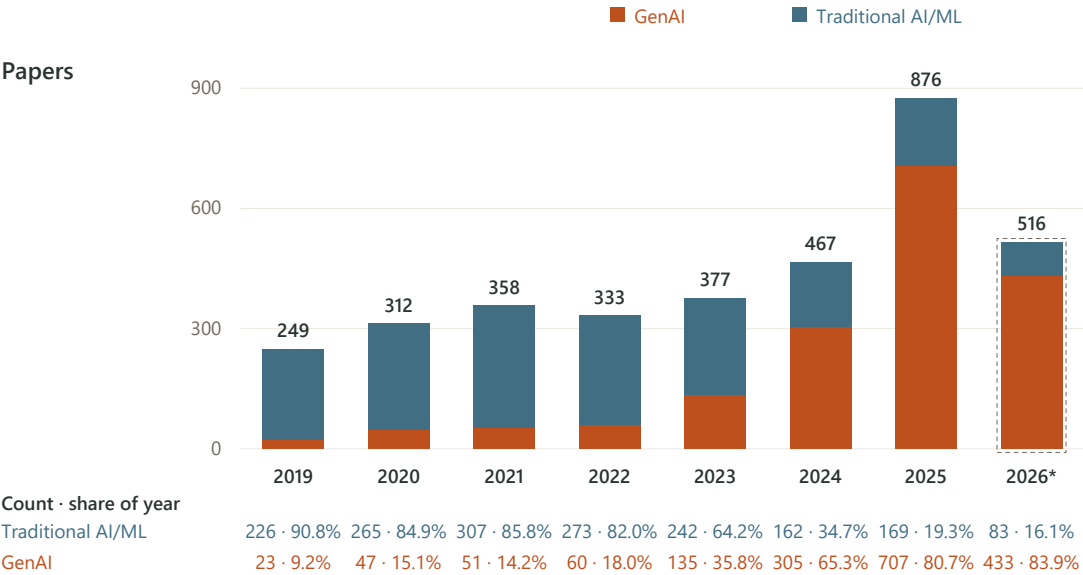


Fig. 55. Research contexts by AI technology. Bars show paper counts across all years; lines show each context's share of its technology group per year, with the coded papers behind those shares below. Contexts overlap. Shares are not adjusted; Table 12 gives the venue-adjusted tests. Only papers with usable context coding and a known technology are included.

AI technologies by year

3,488 classified papers · bar height shows volume; percentages show composition

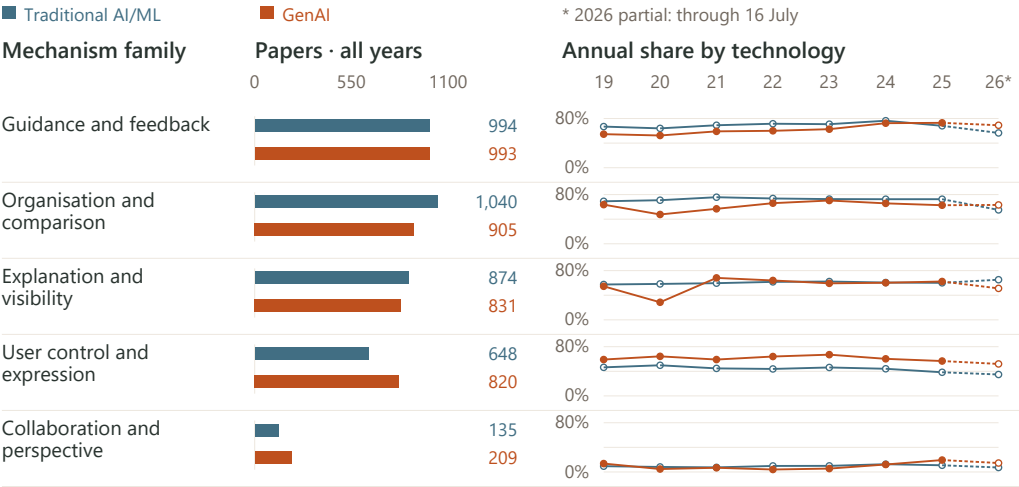


GenAI includes hybrids and earlier generative models. * 2026 through 16 July; September update pending.

Fig. 56. AI technologies by year. Stacked bar heights encode paper counts, with annual totals above. Rows below give each segment's count and within-year percentage. Classification follows the technology used; 2026 is partial (collected in July). The figure omits 645 papers without a resolved technology classification.

Mechanism families: volume and annual share

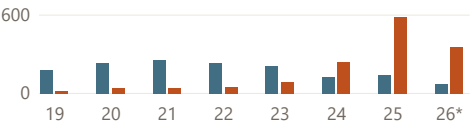
2,880 coded papers with assigned AI technology · categories overlap



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

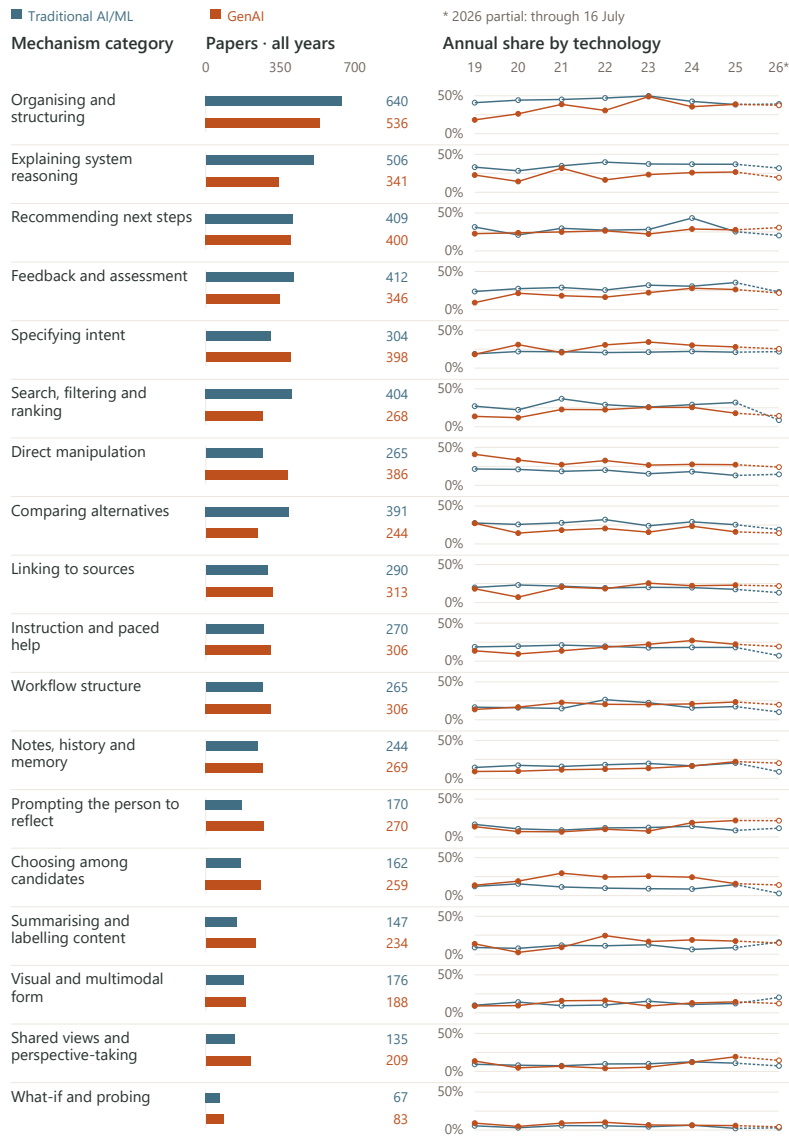


GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 57. Mechanism families by AI technology. Bars encode counts across all years; lines compare annual shares within each technology group. The lower bars show annual coding denominators. Families overlap. Only papers with usable family codes and a known technology are included. Shares are not adjusted; Table 15 gives the venue-adjusted trends.

Mechanism categories: volume and annual share

2,874 coded papers with assigned AI technology - categories overlap



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

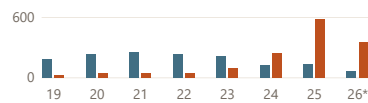


Fig. 58. Mechanism categories by AI technology. Bars encode counts across all years; lines show annual shares within each technology group. Categories overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue. Only papers with usable category codes and a known technology are included.

D.5 Learning overlays

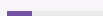
These counts cover the 3,417 papers with usable category coding, of the 4,133 papers in the analysis at 28 September 2026, including papers without an assigned AI technology. Learning scope means the paper is in the learning review; measured means the paper measured learning explicitly. Learning scope and Measured are counted separately within each category; missing or conflicting labels are omitted from the corresponding column.

Table 17. Papers per mechanism category: all papers, papers in learning scope, and papers that measured learning. Counts are distinct papers; categories overlap.

Category	Papers	Learning scope	Measured
Explaining system reasoning	976	213	97
Linking to sources	683	119	34
Summarising and labelling content	423	107	38
Visual and multimodal form	469	155	55
Organising and structuring	1,382	293	94
Comparing alternatives	731	166	61
Search, filtering and ranking	776	177	71
Notes, history and memory	612	170	68
Specifying intent	820	161	56
Direct manipulation	744	138	56
Choosing among candidates	487	104	37
What-if and probing	171	79	32
Recommending next steps	1,000	330	137
Feedback and assessment	906	318	122
Instruction and paced help	688	347	156
Workflow structure	660	173	70
Prompting the person to reflect	506	242	91
Shared views and perspective-taking	410	163	58

D.6 Venue candidates and current inclusions

Table 18. Venue candidates and AI-involved inclusions in the 16 July 2026 frame, assessed through 28 September 2026. The 37,978 venue-year records represent 37,660 candidate identities. Inclusion (%) is Included divided by Initial candidates. Papers without AI, and papers whose identity or scope is unresolved, are not in Included. Venue memberships can overlap; group and overall totals count each identity once.

Venue	Initial candidates	Included	Inclusion%
HCI & interaction	17,436	2,615	 15.0%
56.1% of included corpus			
CHI	7,639	1,152	 15.1%
UIST	860	195	 22.7%

Venue	Initial candidates	Included	Inclusion%
IUI	687	321	46.7%
DIS	1,471	167	11.4%
CSCW	2,412	212	8.8%
C&C	301	44	14.6%
TOCHI	478	38	7.9%
TiiS	230	95	41.3%
EICS	174	23	13.2%
HAI	214	26	12.1%
ECSCW	205	4	2.0%
GROUP	142	6	4.2%
ASSETS	387	58	15.0%
HRI	630	96	15.2%
IVA	222	39	17.6%
CHIIR	312	40	12.8%
HCOMP	112	28	25.0%
IJHCS	960	71	7.4%
Design & creativity	2,502	102	4.1%
2.2% of included corpus			
ICCC	422	28	6.6%
Design Science	232	15	6.5%
Design Studies	234	1	0.4%
IDETC	1,614	58	3.6%
Visualization	6,336	910	14.4%
19.5% of included corpus			
VIS	1,772	352	19.9%
EuroVis	537	84	15.6%
TVCG	4,073	460	11.3%
VL/HCC	255	65	25.5%
Graphics	2,385	137	5.7%
2.9% of included corpus			
SIGGRAPH	1,673	104	6.2%
SIGGRAPH Asia	712	33	4.6%
Learning & education	2,834	228	8.0%
4.9% of included corpus			
L@S	181	34	18.8%
LAK	655	87	13.3%
AIED	1,168	41	3.5%
IJAIED	347	34	9.8%

Venue	Initial candidates	Included	Inclusion%
EDM	483	32	6.6%
Software engineering	3,944	354	9.0%
7.6% of included corpus			
CHASE	249	18	7.2%
ICSE	1,038	129	12.4%
FSE	1,297	88	6.8%
ASE	1,360	119	8.8%
AI & language	2,223	319	14.3%
6.8% of included corpus			
ACL	731	133	18.2%
EMNLP	794	146	18.4%
RecSys	698	40	5.7%
Overall (distinct identities)	37,660	4,665	12.4%

E VOCABULARY GROWTH DURING READING

These curves show how each vocabulary grew as cards were read, so a reader can judge how complete the codes are. The vocabulary built from the first 3,295 mechanism cards gained 12 codes across the next 1,754 papers, and each new code was assigned to an existing category and family. All open fields were checked on 821 of those papers; the other batches checked mechanism facets, values, and domains (Figure 59).

E.1 Reading the Curves

We count distinct values after each batch of ten cards, in reading order, and scale each curve to its own final vocabulary. Codes, categories, and families are drawn where those levels exist; categories and families were assigned after coding, so they can reach their final size early in a curve. Reference lists in the lineage cards are open-ended, while their relation options are fixed by the card format. A flat segment means that no new label appeared in those cards, not that unread papers would add none.

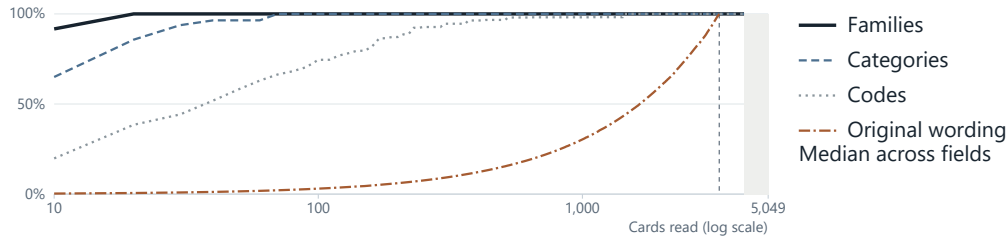
E.2 Stream 1 mechanism cards: Interface and Outcome Fields

These grids show the variation the overview hides. Every panel states its field and final count, and lines are drawn only for the levels present in that field.

Vocabulary growth across three card types

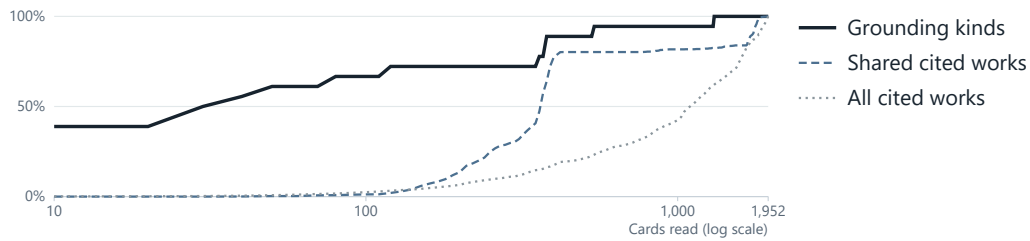
Share of the recorded vocabulary encountered; paper counts use a log scale

Stream 1 · Mechanism cards · 5,049



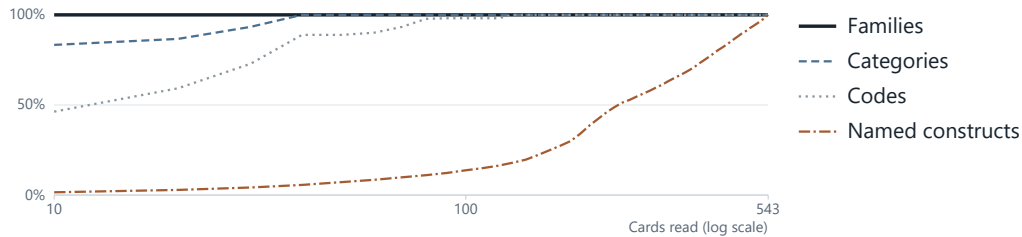
After 3,295: 12 new codes in 1,754 cards; field coverage varies in the shaded region.

Stream 1 · Theory-lineage cards · 1,952



Cited works remain an open vocabulary; grounding kinds are a fixed schema.

Stream 2 · Knowledge-claim cards · 543



Frozen coding covers 542 cards; named constructs cover all 543. Curves stop at their coverage.

Fig. 59. **Vocabulary growth across the three card types.** Panels count cards in reading order on logarithmic axes: 5,049 mechanism cards, 1,952 theory-lineage cards, and 543 Stream 2 cards. The Stream 1 panels include cards of non-AI comparison papers. Shading marks uneven field coverage, and median curves stop before it.

Codes, categories and families reached, by cards read, every field



Fig. 60. **Stream 1, mechanism cards: vocabulary growth in each open field.** Each panel separates codes, categories, and families where present, in reading order.

E.3 Stream 1 theory-lineage cards: Cited Sources and Their Roles

The curve covers the 1,952 newly written lineage cards; reused cards appear in the gallery but not in this sequence.

Stream 1 theory-lineage cards: references and relations

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Relations, positions and grounding kinds use fixed options: their flat tails are not evidence of open-code saturation.

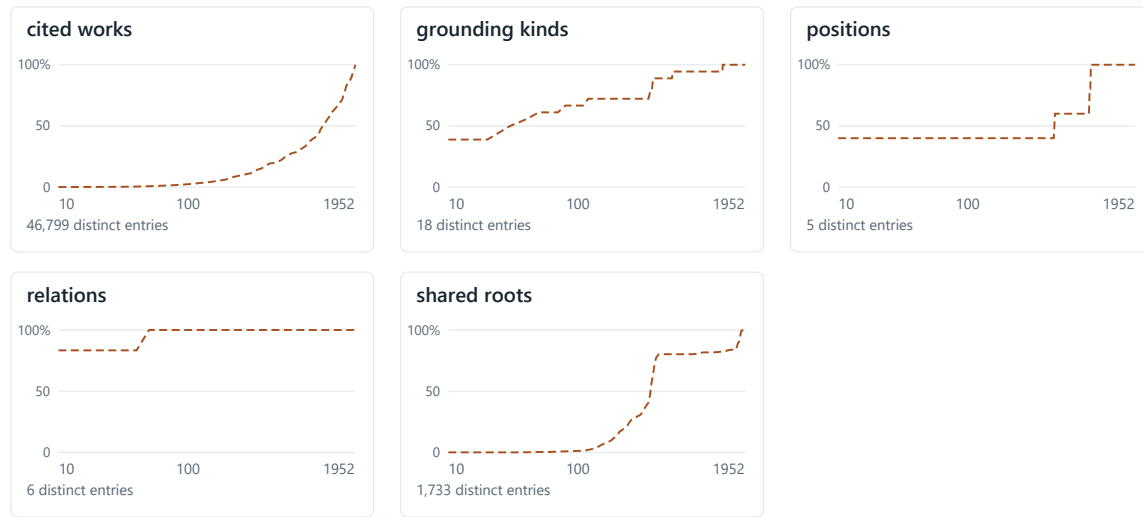


Fig. 61. **Stream 1, theory-lineage cards: reference lists keep growing after relation options are exhausted.** Fixed options and open-ended lists are separate panels because they close in different ways.

E.4 Stream 2: Claims and Conditions

Stream 2: each coded field

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Dotted: raw values. Dashed: canonical codes. Solid: categories. Thick: families, where present.

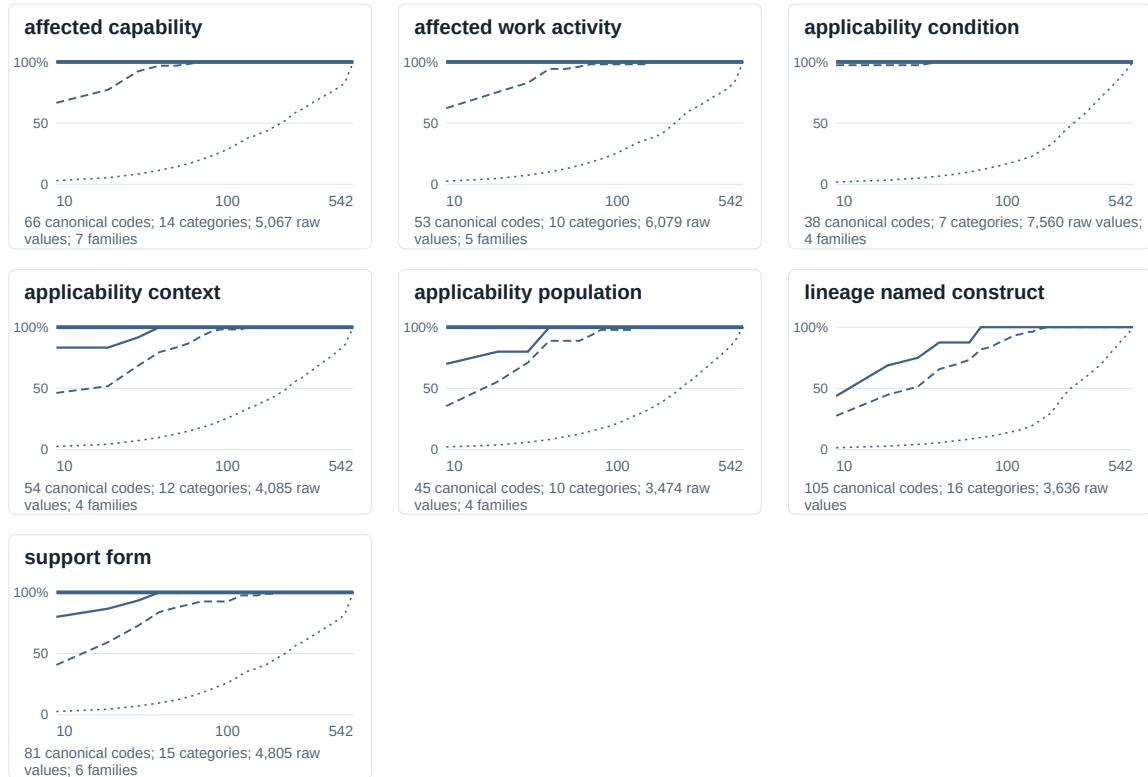


Fig. 62. **Stream 2: vocabulary growth differs by coded field.** Codes, categories, and families are drawn only where that level exists.

Stream 2's seven open fields cover 542 cards and 35,699 observations. Of the 449 codes, 448 are used, and no new code entered in the last 16 coding batches. A flat curve can still hide a missing code: no population code names a studied unit that is or includes a machine, and 18 assignments flag this while using codes for people or artifacts.

E.5 New Codes Within the First Coding

Within the first coding of each open field, the last new code appeared between 2.5% and 48.7% of the way through the fields whose coding order was kept (Figure 63). Three fields still have low-confidence assignments for more than a quarter of their observations, so a vocabulary can stop growing and remain too coarse.

Where did the recorded vocabulary stop expanding?

Colour ends at the last new code. The pale remainder reuses codes already encountered.



■ Nine fields retain observation order; ticks show first-use counts within each tenth of that sequence.

■ Twelve fields retain batch order only; their bars locate the last batch that introduced a code.

This checks label reuse within the recorded sequence, not coverage of every meaningful distinction. Stable labels can still contain low-confidence or overly broad assignments.

Fig. 63. **New-code creation within each field of the first coding.** Bars locate the last new code in the recorded coding sequence; ticks show new codes by tenth of the sequence. Where only batch order survives, the figure reports batch resolution.